

Naming Elementary Embeddings

Atom Components, Replacement,
and Reflection

Exact criteria and sharp cofinality thresholds

Prepared for Vladimir Reshetnikov

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Research status. Complete mathematical arguments are supplied for the stated results. The kernel-ideal construction is established prior work, explicitly credited below. The component-profile criterion and its finite-signature refinements are proposed contributions of this manuscript; independent review and literature-priority verification remain outstanding. No Lean verification is claimed. Elliot Glazer and Bokai Yao are cited researchers, not authors or endorsers of this manuscript.

[write] Filed in Provelt's research-report collection as `ordinals-and-order-types/naming-elementary-embeddings` (batch 89, manuscript 05; label prefix `nee:`). Provenance, formal status, the relation to the repository and the notation table are in Sections 1.4 and 1.5.

[write] Since 4 October 2026 the report has two Parts: Part I (Sections 1–11 and Appendices A–C; batch 89, labels `nee:`) is the original report, with its numbering unchanged; Part II (from Section 12; batch 93, labels `nee:ca:`) treats commuting actions and uniformly named groups, answers Question 10.4 of Part I, and answers Question 10.1 for commuting permutations and the fixed-action part of Question 10.10. The abstract below is Part I’s; Part II has its own. One bibliography serves both Parts.

Abstract

Let A be an external set of atoms, let κ be an infinite cardinal with $|A| \geq \kappa$, and let $M_\kappa(A)$ consist of the objects whose transitive closures involve fewer than κ atoms. This is a standard kernel-ideal model of replacement-based set theory with urelements and choice. Every injection $f : A \rightarrow A$ has a canonical, amenable elementary lift $j_f : M_\kappa(A) \rightarrow M_\kappa(A)$ fixing every pure set. Nevertheless, naming these lifts can invalidate Replacement.

For a finite tuple f of atom injections, consider the connected components of its labeled functional graph. Write B_t for the union of all components of isomorphism type t , and put

$$D_\kappa(f) = \bigcup \{B_t : |B_t| < \kappa\}.$$

We prove that for $\kappa > \omega$, Replacement in the expanded language holds exactly when $|D_\kappa(f)| < \kappa$. For $\kappa = \omega$, it holds exactly when this union is finite and all components are finite. The proof combines uniform internal coding of component types with a fresh-component automorphism argument.

There are countably many one-injection component types, but continuum many types for two injections, already for two permutations. Consequently every one-symbol expansion preserves Replacement exactly when $\text{cf}(\kappa) > \omega$, whereas every expansion by $r \geq 2$ fixed, finitely many symbols preserves it exactly when $\text{cf}(\kappa) > 2^{\aleph_0}$. At the cutoff $\aleph_2$, the latter universal preservation property is equivalent, externally, to the continuum hypothesis. This is a characterization, not a resolution, of that hypothesis.

We also classify the initial elementary self-embeddings, determine precisely which canonical lifts are parameter-definable, and prove that full Collection and full transitive reflection in an expanded model hold exactly when κ is a successor cardinal and fewer than κ component types are realized. At limit cutoffs, the exact domain-size threshold for Collection is $\text{cf}(\kappa)$, with failure already in the unexpanded bounded formula fragment. The final sections identify concrete interfaces with Prover’s first-order set theory and bounded-consistency developments and propose twelve further research directions.

Keywords. Urelements; elementary self-embeddings; amenable classes; expanded-language Replacement; Collection; reflection; component types; continuum hypothesis; formalization.

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Part I

Atom components, Replacement, and reflection

1 Motivation, scope, and the contribution boundary

1.1 Why this intersection is substantive

Glazer and Yao's *Reflection Principles in ZFU* studies how Collection and several forms of reflection diverge when urelements are admitted, especially without choice [1]. Glazer's work on global choice over Zermelo set theory emphasizes another source of strength: adjoining a class function and allowing its symbol in an axiom scheme is not merely a change of notation [2]. These two themes meet in a concrete question:

When an elementary self-embedding is named, which set-existence schemes continue to hold in the expanded language?

ProveIt supplies a complementary perspective. Its first-order ZF development represents Separation and Replacement as constructions on formulas, while its bounded-consistency project explicitly distinguishes external families of formulas from single internally quantified assertions [4, 5]. A new symbol changes the formulas over which a scheme ranges. This manuscript studies that change semantically in models where the obstruction can be computed exactly.

The repository also contains a substantial, explicitly unformalized report on surreal self-embeddings [6]. Here the embeddings are of a set-theoretic universe with atoms, not of the surreal ordered field. They fix the entire pure universe. Thus they provide a useful control example: nontrivial elementary self-embeddings need not move ordinals when atoms are present, but they do not produce nontrivial embeddings of the usual pure-set presentations of surreal numbers.

1.2 The mathematical question

An elementary map preserves every formula of a specified language. An amenable map has a set-sized restriction to each internal set. Neither statement says that *every function definable using a name for the map has an internal range set*. That third assertion is expanded-language Replacement.

The difference is not cosmetic. We will construct an automorphism j for which every restriction $j \upharpoonright x$ belongs to the model, yet Replacement fails once j is named. More sharply, at the cutoff $\aleph_1$ we construct two automorphisms such that each one-symbol expansion satisfies Replacement, Collection, and reflection, while their joint expansion fails Replacement.

The main classification is local to a precisely specified family of models. It is not a classification of all models of set theory with urelements, all amenable classes, or all elementary embeddings of the universe.

1.3 What is known and what is proposed here

The construction using an ideal of atom kernels is not new. Yao's Definition 24 and Theorem 26 give it in substantially greater generality; his Theorem 27 contains relevant finite, countable, and

larger kernel-bound examples [3]. We reproduce the elementary symmetry argument needed for the later results, rather than treating that construction as an original discovery.

Part of the manuscript	Status and role
Kernel-ideal models	Established construction, credited to Yao; reviewed for self-containment.
Canonical atom lifts	Explicit structural development using standard permutation symmetry; no claim that the underlying lifting idea is new.
Component-profile criterion	Principal proposed contribution: an exact criterion for full expanded-language Replacement.
One-name versus two-name thresholds	Derived proposed contribution, including an explicit coding construction and a CH characterization.
Collection and reflection spectrum	A detailed calibration for the expanded family; the unexpanded cutoff phenomena build on the established kernel-ideal method.
Formalization	A concrete design and dependency audit, not a completed Lean development.

No assertion below is advertised as the solution of an open problem explicitly posed by Glazer. Instead, the paper formulates and completely answers the model-family classification questions above. The available literature search establishes the prior-art boundary just described; it does not establish publication priority for every refinement.

1.4 Provenance and place in the collection

[write] Added 3 October 2026, batch 89. This report is a single manuscript, printed in full: manuscript 05 of batch 89 of Provelt’s incoming-reports intake, delivered as the archive `Naming_Elementary_Embeddings_Research.zip` (330,868 bytes; inner directory `Naming_Elementary_Embeddings/`, main file `article.tex`, 1041 lines, a 27-page US Letter PDF) in the arrival commit `8ea27d6c0`, and placed in commit `23adb85f9` in the research-report collection as `ordinals-and-order-types/naming-elementary-embeddings`. It contains no surreal mathematics and continues the formal projects `SetTheory/ZF` and `SetTheory/BoundedConsistency`, so it was filed in the collection, in the category that already holds Provelt’s other set-theoretic reports, rather than in either project. The author lines read “Prepared for Vladimir Reshetnikov” and “Research manuscript prepared with ChatGPT”; the report is AI-assisted, unrefereed and not formalized. Its four sister manuscripts of batch 89 became additions to surreal-collection reports: Part XV of `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders` (manuscript 01), Part X of `.../polish-models-of-omnific-arithmetic` (02), and Parts VI and VII of `.../birthday-cutoffs-and-hereditary-sets` (03 and 04). Only manuscript 03 shares a theorem with this report (see below). Being a single source, the report needed no merge choices. The write prefixed all 64 delivered labels with `nee:` and updated every reference (`38 \cref keys`, `13 \eqref`); added the two labels `nee:sec:provenance` and `nee:sec:notation`; added these dated **[write]** notes, the title-page line and the `pageanchor` switch around the title page (a delivered build gave a duplicate `page.1` destination). No statement, proof, number or non-claim of the manuscript was changed; no section, theorem or equation number moved. [Added 4 October 2026, batch 93: this paragraph describes what is now Part I. Batch-93 manuscript 01, *Commuting Actions and Replacement*, written against this report, became Part II (Section 12.4 gives its provenance); wrapping the text above as Part I moved no number, and the single source of Part I still needed no merge choices.]

Pin and repository claims. The pin `2b7b388ba81a3355b19a7c2d2fe797e92f572355` (Section 9 and the repository entries of the bibliography) is the Provelt commit “Catalogue batches 81 to 86” of 3 October 2026, an ancestor of the placement. The three repository files the manuscript read are byte-identical at the pin and

at the write: SetTheory/ZF/Lean/ZF/Zf.lean (blob b33db57cf), SetTheory/BoundedConsistency/README.md (ebb5d1bfa) and Algebra/SurrealNumbers/docs/surreal/surreal-self-embeddings/README.md (554357db9). What Sections 1, 8 and 9 say about them is accurate: Zf.lean declares Sep_form, Func_form, Image_form, Repl_form, ZFax, ZFprov and ZFax_s, with schema constructors over the formula type Form, semantic bridges and an internal finite-recursion theorem; the bounded-consistency README states a numeralwise endpoint, chooses partial satisfaction over Levy reflection, and records that its quantifier-group rank, in a syntax without primitive bounded quantifiers, is strictly finer than the Levy hierarchy; the self-embedding report is marked unrefereed and not formalized. No repository statement is contradicted, and nothing needed retraction.

Formal status. No statement of this report is formalized in Lean or Rocq, and its place beside the formal projects confers no formal status. What those projects do formalize is pure membership set theory: SetTheory/ZF/Lean/ZF/Zf.lean (a Lean port of the Rocq file SetTheory/ZF/Coq/Zf.v, which declares Sep_form, Repl_form, ZFax and ZFax_s likewise) proves the bridges bridge_Sep and bridge_Repl and, inside every model of Extensionality, Separation, Pairing, Union, Infinity and Replacement, the closure theorem ClosureF0_of_ZF; SetTheory/BoundedConsistency proves, by partial satisfaction, the numeralwise endpoint BoundedZFConsistency.Endpoint.zfc_proves_conZFC ($\text{ZFC} \vdash \text{Con}_n(\text{ZFC})$ for each metatheoretic n). Their formula type Form (Logic/FirstOrder/Lean/FirstOrder/Fol.lean) is purely relational, with the atomic formulas $\in$ and $=$ only. Neither the base language $\mathcal{L}_U = \{\in, \text{At}\}$ nor the expanded language $\mathcal{L}_r$ of this report exists there, so the “first task” of Section 9 and all six layers that follow it are outstanding. No urelement theory is formalized anywhere in Provel.

Review in the Hilbert’s-tenth research tree. Before placement, another session routed all five batch-89 archives (commit 6aadaa4f2): Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_topology_surreal_arrivals_20261003.md, with the receipt review_topology_surreal_arrivals_20261003.json beside it. For this manuscript it read the delivered README and source audit, the recorded finite-check scope, and the delivered lines 118–133, 183–211, 807–865, 911–927 and 988–994 of article.tex (the abstract, Section 2 through Remark 2.1, Section 9, the last two questions of Section 10 with the conclusion, and Appendix C). It records that “finitely many names” is a language parameter, not a bound on Diophantine witnesses, that the finite checks are regression evidence, not verification of the infinite theorems, and that the proposed formalization is not an implemented compiler; it finds nothing that changes the Hilbert’s-tenth programme’s 84-operation universal polynomial. It claims no theorem audit, literature check or replay of the finite checks, and it made no correctness finding; nothing here needed a change of scope.

The same Collection theorem in a surreal report. Batch-89 manuscript 03, *Surreal Cuts, Replacement, and Atom-Support Spectra*, written against the same pin and placed in the same commit, is Part VI of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (labels hset:zr:; written concurrently with this report). It studies the same kernel models, which it writes M_κ^{ker} (manuscript 03’s M_κ , renamed there), and proves, in its delivered numbering (Part VI numbering in brackets, added 4 October 2026, after that write was committed): kernel confinement of a unique output (its Lemma 9.1 [Lemma 32.1, hset:zr:lem:confinement], our lemma 2.3); the base model theorem (its Theorem 9.2 [Theorem 32.2, hset:zr:thm:supportbase], our theorem 2.4; this report credits the construction to Yao [3], manuscript 03 to Yao and to Hamkins–Yao with Lévy); and the Collection spectrum of these models (its Lemma 10.1, Theorems 10.2–10.3 and Corollary 10.4 [Lemma 33.1, Theorems 33.2–33.3 and Corollary 33.4, hset:zr:cor:classification]): Collection holds on domains of size below $\text{cf } \kappa$ for every infinite κ , fails on a domain of size $\text{cf } \kappa$ when κ is a limit cardinal (ω included), and holds in full when κ is a successor. That is the unexpanded case of lemma 7.1, lemma 7.2, corollary 7.3 and theorem 7.5 (take $f = (\text{id}_A)$, so that $|I(f)| = 1$). The two proofs are independent and both stand; neither manuscript cites the other, and neither claims priority for the unexpanded spectrum. They differ as the note after corollary 7.6 records. Part VI records the overlap in its Remark 24.6 (hset:zr:rem:nee) and prints manuscript 03’s own proofs. Part XV of surreal-well-orders (manuscript 01) finds the same cofinality threshold $\text{cf } \lambda$ for class Replacement in the full-class rank models $(V_\lambda, \mathcal{P}(V_\lambda))$; those are different models, and no proof is shared.

Neighbouring reports. The surreal report birthday-cutoffs-and-hereditary-sets also proves, in its subsection

“Transfer of Kunen’s inconsistency” (Theorem `hset : thm : kunen`), that the birthday-expanded surreal class has no nontrivial class elementary self-embedding, under the Kunen hypotheses that allow the embedding as a class parameter in Separation and Replacement. This report shows how strong that hypothesis is in another setting: in the urelement models $M_\kappa(A)$ nontrivial amenable elementary self-embeddings exist (theorem 3.2), and naming them may or may not preserve Replacement (theorem 5.1). There is no conflict: every j_f fixes the whole pure universe, and Kunen’s theorem concerns embeddings of a pure universe with choice. The self-embedding report cited as [6] makes the analogous point in its Remark `sse : pf : rem : kunen`. In this category, ordinals-and-order-types/measurable-box-games also answers a Glazer paper and also cites [2], but shares no theorem or notation, and ordinals-and-order-types/lexicographic-well-orderings-of-reals concerns pure ZFC. Apart from Parts VI and VII of birthday-cutoffs-and-hereditary-sets (manuscripts 03 and 04; Part VII’s Fraenkel–Mostowski permutation models are choiceless models with atoms, unlike the models here), no report in ProveIt treats urelements or cites [1] or [3]. [Added 4 October 2026, batch 90: Part VIII of the same surreal report (batch-90 manuscript 05) also works in finite-kernel models with atoms and cites [1] (see the note at the end of section 5), and Part XII of polish-models-of-omnific-arithmetic (batch-90 manuscript 03) cites [1] as context for its class theory without Foundation; neither cites [3].]

The cited papers, checked at the write. On 3 October 2026 the arXiv record of [1] listed a single version, v1 of 25 February 2026 (math.LO), with the cited title and authors; the record of [3] lists three versions, the last (v3, the version cited) of 18 June 2023; the record of [2] lists three versions, the last of 7 January 2024. The page references to Yao’s Definition 24 and Theorems 26–27 were not rechecked.

1.5 Notation used in this report

[write] Added 3 October 2026, batch 89. No symbol was renamed. Several symbols of this report occur with other meanings in neighbouring reports; this table fixes the readings here, with the tempting false one where confusion is likely.

Symbol	Meaning here (and what it is not)
$M_\kappa(A)$	the kernel-cutoff model (2.1); Part VI of birthday-cutoffs-and-hereditary-sets writes M_κ^{ker} for the same class (manuscript 03’s M_κ). <i>Not</i> the full-class rank model $M_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$ of Part XV of surreal-well-orders, nor the Presburger models M_α of polish-models-of-omnific-arithmetic
$\mathcal{N}_{\kappa, f}$	the expansion of $M_\kappa(A)$ by the named lifts; not the natural numbers
$\ker(x)$	the atom kernel $A \cap \text{TC}(\{x\})$; not the kernel of a homomorphism
$\text{mov}(f)$	the atoms moved by f (theorem 3.6); not a support in the sense of permutation models
$\text{ZFCU}_R, \text{ZFU}_R$	urelement set theory with Replacement and without Collection; the subscript R names Replacement, not the reals
$B_t, D_\kappa(f), I(f)$	the type block of t , the union of the small blocks, the set of realized types. B_t is not the birthday-cutoff structure B_κ of birthday-cutoffs-and-hereditary-sets; the B_n, B_i of the examples are atom blocks
H	in (5.4) and lemmas 5.2 and 7.4, a protected union of components of size below κ ; not the hereditary-size universe H_κ , and not manuscript 03’s hierarchy axiom H
$V, V_\alpha(C)$	the pure universe (Example 3.8); the cumulative hierarchy over the atom set C
$\text{Collection}_{\leq \mu}(J)$	Collection on domains of size <i>at most</i> μ ; manuscript 03 writes $\text{Collection}_{< \tau}$ for size <i>below</i> τ
$\text{Rob}_r(\kappa, A), \text{RP}(J)$	universal Replacement preservation for r names (Section 6); full transitive reflection in the expanded language
$\mathfrak{c}$	$2^{\aleph_0}$
limit cardinal	includes ω (lemma 7.2); “successor” in theorem 7.5 means an uncountable successor

2 Ambient foundations and the kernel-cutoff model

2.1 Two universes must not be confused

Work in an external, well-founded set universe satisfying choice, with a *set* A of atoms. The full cumulative universe of sets over A is denoted by $U(A)$. This setup can be implemented in ordinary ZFC using tagged atoms and tagged sets; membership in the coded universe is then the interpreted membership relation. No inaccessible cardinal or external proper class of atoms is required. Whenever witnesses are chosen below, external Collection first bounds them inside a set; reservoir choices and component isomorphisms are also choices from sets. No global class-choice principle is assumed.

An atom has no members and is distinguished from the empty set by a unary predicate At . Put $\text{Set}(x) \equiv \neg \text{At}(x)$. Extensionality is restricted to sets. Define

$$\ker(x) = \{a \in A : a \in \text{TC}(\{x\})\}.$$

In particular, $\ker(a) = \{a\}$ for an atom a . An object is *pure* when its kernel is empty. The pure part is a copy of the full external pure-set universe.

Fix throughout an infinite cardinal κ and a set A satisfying $|A| \geq \kappa$. Our model is the definable external class

$$M_\kappa(A) = \{x \in U(A) : |\ker(x)| < \kappa\}. \quad (2.1)$$

All cardinal inequalities in the definition are evaluated externally. Every individual atom belongs to $M_\kappa(A)$, but A itself is not a set of the model. This distinction is the engine of the examples.

For any external $C \subseteq A$, let $V_\alpha(C)$ be the usual cumulative hierarchy over C :

$$V_0(C) = C, \quad V_{\alpha+1}(C) = \mathcal{P}(V_\alpha(C)) \cup C, \quad V_\lambda(C) = \bigcup_{\alpha < \lambda} V_\alpha(C).$$

Every object whose kernel is contained in C belongs to some $V_\alpha(C)$. Each $V_\alpha(C)$ is an external set, transitive in the urelement sense, and has kernel contained in C .

Remark 2.1 (Schemas, not a truth predicate). For a fixed first-order formula φ , the assertion $M_\kappa(A) \models \varphi$ means its external relativization to the definable class (2.1). We use one such relativization at a time. We do not postulate a first-order predicate uniformly defining truth for all formulas of a proper-class structure. Claims of elementarity and reflection are corresponding metatheoretic schemes. Quantification over externally given class embeddings is likewise metatheoretic, not an internal set of all class maps.

2.2 The base axioms and internal cardinalities

Let ZFCU_R denote set theory with atoms, Extensionality for sets, Foundation, Pairing, Union, Powerset, Infinity, Separation, Replacement, and the assertion that every set can be well-ordered. Collection is *not* included as an axiom.

Lemma 2.2 (Kernel arithmetic). *For sets and objects in $U(A)$, the following hold:*

1. If $y \in x$, then $\ker(y) \subseteq \ker(x)$.
2. $\ker(\{x, y\}) = \ker(x) \cup \ker(y)$.

3. If x is a set, then $\ker(\bigcup x) \subseteq \ker(x)$ and $\ker(\mathcal{P}(x)) \subseteq \ker(x)$.
4. If Y is a set of objects, then $\ker(Y) = \bigcup_{y \in Y} \ker(y)$.

Consequently $M_\kappa(A)$ is transitive, contains every pure set, and is closed under subsets of its sets and under the ordinary finite set operations.

Proof. Membership chains witnessing occurrence of an atom concatenate through an element of a set. This proves the first, second, and fourth assertions. A membership chain through a subset of x introduces no atom not already occurring through x , giving the powerset assertion; the union assertion is similar. A finite union of cardinals less than an infinite cardinal κ is still less than κ , whether or not κ is regular. $\square$

Any permutation π of A extends recursively to $U(A)$ by

$$\hat{\pi}(a) = \pi(a), \quad \hat{\pi}(x) = \{\hat{\pi}(y) : y \in x\} \quad (\text{Set}(x)).$$

The extension is an automorphism of $M_\kappa(A)$, because it preserves kernel cardinalities. It fixes an object whenever it fixes all atoms in its kernel.

Lemma 2.3 (No new atoms in a unique base-language output). *Suppose $x, p \in M_\kappa(A)$ and*

$$M_\kappa(A) \models \exists! y \varphi(x, y, p),$$

where φ belongs to $\mathcal{L}_U = \{\in, \text{At}\}$. For its unique witness y ,

$$\ker(y) \subseteq \ker(x) \cup \ker(p).$$

Proof. Assume b belongs to the left side but not the right. Since the union of the three kernels has cardinality less than κ , choose an atom c outside that union. The transposition of b and c fixes x, p , preserves the model and its language, and changes the kernel of y . Its lift therefore supplies a different witness to the same formula, contradicting uniqueness. $\square$

Theorem 2.4 (Known kernel-ideal theorem, specialized). $M_\kappa(A) \models \text{ZFCU}_R$. *Its class of atoms is internally proper. It has the same pure ordinals and pure cardinal arithmetic as the ambient pure universe. For each internal set x , its cardinality computed internally equals its external cardinality.*

Proof. The structural axioms and Separation follow from [lemma 2.2](#). Given a functional formula on a set w , external Replacement produces its range Y . By [lemma 2.3](#),

$$\ker(Y) \subseteq \ker(w) \cup \ker(p),$$

so Y is internal. This proves every instance of base-language Replacement.

An external well-order of x is a subset of $x \times x$ with kernel contained in $\ker(x)$, hence is internal. Its ordinal enumeration and any external bijection between x and a pure ordinal have the same kernel bound. Thus internal and external cardinalities of x agree, and choice holds internally. All pure objects and their relations are present, so pure cardinal calculations agree as well. Finally, a set containing all atoms would have kernel A , contradicting the cutoff. $\square$

The theorem is a specialization of Yao's ideal construction [[3](#), Definition 24 and Theorem 26]. Notice that the proof bounds the atoms in a uniquely specified output. It does *not* choose witnesses for an arbitrary relation. The latter problem is Collection and behaves differently.

3 Canonical elementary lifts and definability

3.1 Every atom injection gives an elementary map

For any external injection $f : A \rightarrow A$, define its canonical lift by well-founded recursion:

$$j_f(a) = f(a) \quad (a \in A), \quad j_f(x) = \{j_f(y) : y \in x\} \quad (\text{Set}(x)). \quad (3.1)$$

This is an external definable class function with parameter f ; the graph of f need not belong to $M_\kappa(A)$.

Lemma 3.1 (Extending a small partial bijection). *If $P \subseteq A$, $|P| < \kappa$, and $h : P \rightarrow A$ is injective, then h extends to a permutation of A .*

Proof. Both $A \setminus P$ and $A \setminus h[P]$ have cardinality $|A|$, since fewer than $\kappa \leq |A|$ points were removed. Choose a bijection between the complements and combine it with h . $\square$

Theorem 3.2 (Elementary and amenable lifts). *The map j_f restricts to an elementary embedding of $(M_\kappa(A), \in, \text{At})$ into itself. It preserves rank, fixes every pure set, and satisfies*

$$\ker(j_f(x)) = f[\ker(x)]. \quad (3.2)$$

For every internal set x , the graph of $j_f \upharpoonright x$ is internal. Its range as a class is

$$j_f[M_\kappa(A)] = \{x \in U(A) : \ker(x) \subseteq f[A], |\ker(x)| < \kappa\}.$$

Proof. External well-founded induction gives injectivity, preservation of membership and atomhood, rank preservation, and (3.2). The kernel identity shows that the map preserves the cutoff. The same induction shows that it fixes pure objects.

To prove elementarity, fix one formula and a finite tuple $\bar{x}$ of parameters. Let P be the union of their kernels. By lemma 3.1, $f \upharpoonright P$ extends to a permutation π of A . The automorphism $\hat{\pi}$ agrees with j_f on $\bar{x}$. Thus the formula has the same truth value at $\bar{x}$ and at $j_f(\bar{x})$. This is the required elementary-embedding scheme.

External Replacement constructs the restriction graph on x , whose kernel is contained in $\ker(x) \cup f[\ker(x)]$, a set of size less than κ . This proves amenability. Finally, the inverse bijection $f[A] \rightarrow A$ lifts on objects supported on $f[A]$, giving the stated range. $\square$

Definition 3.3. An elementary self-embedding is *initial* here when its range is transitive: if $z \in j(x)$, then z belongs to the range of j .

Theorem 3.4 (Classification of initial embeddings). *Every initial elementary self-embedding of $(M_\kappa(A), \in, \text{At})$ is j_f for a unique injection $f : A \rightarrow A$. Conversely every j_f is initial. Composition corresponds to composition of atom injections.*

Proof. An elementary embedding restricts on atoms to an injection f . If $z \in j(x)$, initiality gives $z = j(y)$ for some y , and preservation and reflection of membership give $y \in x$. Hence, for every set x ,

$$j(x) = \{j(y) : y \in x\}.$$

External well-founded induction identifies j with (3.1). The range description in theorem 3.2 proves the converse. Uniqueness and the composition law follow from the recursion. $\square$

The initiality hypothesis is essential to the scope of this theorem. We do not classify elementary self-embeddings with nontransitive range.

Proposition 3.5 (Elementary subuniverses over smaller atom reservoirs). *For an external $C \subseteq A$, put*

$$M_\kappa(C) = \{x \in U(A) : \ker(x) \subseteq C, |\ker(x)| < \kappa\}.$$

Then $M_\kappa(C) \preccurlyeq M_\kappa(A)$ exactly when $|C| \geq \kappa$.

Proof. Suppose $|C| \geq \kappa$. For parameters supported on $P \subseteq C$ and an existential witness supported on K , move $K \setminus P$ into $C \setminus P$ by a permutation fixing P . The required partial bijection extends because all prescribed supports have size less than κ . The moved witness belongs to $M_\kappa(C)$ and satisfies the same formula. The Tarski–Vaught argument, applied one formula at a time, proves elementarity.

If $|C| < \kappa$, the set C itself is in the smaller model. The formula asserting an atom outside C is true in $M_\kappa(A)$ and false in $M_\kappa(C)$. $\square$

3.2 Exactly which lifts are definable?

Theorem 3.6 (Small movement is equivalent to parameter-definability). *Let $\text{mov}(f) = \{a \in A : f(a) \neq a\}$. The canonical lift j_f is definable in the base-language structure $M_\kappa(A)$ from an internal parameter if and only if*

$$|\text{mov}(f)| < \kappa. \quad (3.3)$$

If p is a parameter defining j_f , then $\text{mov}(f) \subseteq \ker(p)$.

Proof. Suppose p defines j_f and put $P = \ker(p)$. Every atom permutation fixing P pointwise preserves the definition, so f commutes with every such permutation. For $a \notin P$, a permutation fixing $P \cup \{a\}$ can move any atom outside $P \cup \{a\}$. Therefore $f(a) \in P \cup \{a\}$. If $f(a) \in P$, interchange a with a different $b \notin P$. Commutation gives $f(b) = f(a)$, contrary to injectivity. Thus $f(a) = a$ outside P .

Conversely put $S = \text{mov}(f)$ and assume $|S| < \kappa$. Injectivity implies $f[S] \subseteq S$: an image outside S would collide with that outside atom's fixed image. The graph $g = f \upharpoonright S$ is an internal set. It defines f on atoms, with identity elsewhere.

There is a uniform first-order definition of the recursive lift from g . For clarity, on $D = \text{TC}(\{x\})$ ask for a set function h that uses the prescribed atom map and satisfies

$$h(z) = \{h(u) : u \in z\} \quad \text{for each set } z \in D.$$

External well-founded recursion supplies the unique such function. Its graph and every value have kernels contained in $\ker(x) \cup S$, hence are internal. The displayed recursion can be expressed by ordinary first-order quantification over the graph h . It therefore defines $j_f(x)$ internally from g , uniformly in x . $\square$

Corollary 3.7. *The only parameter-free canonical self-embedding is the identity. At $\kappa = \omega$, every parameter-definable initial elementary self-embedding is an automorphism. At every $\kappa > \omega$, there is a parameter-definable proper initial elementary self-embedding. If j_f is defined by p and $j_f(p) = p$, then j_f is an automorphism.*

Proof. A definition without parameters is invariant under all atom permutations, so the proof above has $P = \emptyset$. Finite movement makes an injection a finite permutation, giving the ω case. For $\kappa > \omega$, shift an internal countably infinite set of atoms forward by one and fix its complement; the moved set is countable and the map is not onto.

For the last assertion, (3.2) gives $f[P] = P$ when $j_f(p) = p$. Since f fixes $A \setminus P$ and maps P onto itself, it is surjective. $\square$

Example 3.8 (An elementary map with exactly the pure fixed points). Identify A externally with $B \times \omega$ and let $f(b, n) = (b, n + 1)$. A nonempty subset of A cannot satisfy $f[S] = S$: in a nonempty fiber choose its least second coordinate. If $j_f(x) = x$, (3.2) forces the kernel to be empty. Thus

$$\text{Fix}(j_f) = V,$$

where V is the pure universe. Also $\bigcap_{n < \omega} j_f^n[M_\kappa(A)] = V$. Each range is elementary by [proposition 3.5](#), whereas their intersection is not elementary because it has no atoms. Every ordinal is fixed; there is no critical ordinal and no large-cardinal conclusion.

4 Expanded languages and countable component types

4.1 What naming a map asserts—and what it does not

Fix a positive integer r and external injections $f = (f_1, \dots, f_r)$ on A . Write

$$\mathcal{L}_r = \{\in, \text{At}, J_1, \dots, J_r\}, \quad \mathcal{N}_{\kappa, f} = (M_\kappa(A), \in, \text{At}, j_{f_1}, \dots, j_{f_r}).$$

Here J_i is a unary function symbol. We write $\text{Replacement}(J)$ and $\text{Collection}(J)$ for the schemes whose formulas may contain any of these symbols. The notation does not mean only the special instance asserting the existence of $J_i''x$.

Every instance of expanded Separation holds: externally separate a subset of the given internal set, then use the kernel bound. All the other base axioms except potentially expanded Replacement remain true for the same reason or because they have no new schema instances. Every named map is amenable by [theorem 3.2](#).

If an atom permutation π commutes with every f_i , its lift is an automorphism of $\mathcal{N}_{\kappa, f}$. This follows first on atoms and then by recursion on sets. Such simultaneous symmetries, rather than arbitrary atom permutations, control expanded Replacement.

4.2 Functional graphs and their pure codes

View $(A; f_1, \dots, f_r)$ as a labeled directed graph, with an edge labeled i from a to $f_i(a)$. A *component* means a connected component after directions are forgotten. There are at most $2r$ neighbors at a vertex, since each f_i is injective. Thus every component is finite or countably infinite.

Let $\mathcal{C}_r$ be the pure set of codes for connected structures with r total injections and domain either a nonzero natural number or ω . Isomorphism respects every function label and direction. Let $\mathcal{T}_r$ be the quotient of $\mathcal{C}_r$ by isomorphism. An element of $\mathcal{T}_r$ is taken literally as a set of codes in $\mathcal{C}_r$; this avoids a choice of canonical representatives. All these objects are pure sets and therefore belong to $M_\kappa(A)$.

For $t \in \mathcal{T}_r$, define externally

$$B_t = \{a \in A : \text{the component of } a \text{ has type } t\}, \quad I(f) = \{t \in \mathcal{T}_r : B_t \neq \emptyset\}. \quad (4.1)$$

The B_t are pairwise disjoint. They need not be internal sets; $I(f)$, being pure, always is.

Lemma 4.1 (Number of possible component types). *For one injection, $\mathcal{T}_1$ is countably infinite. Its types are finite directed cycles, a one-sided ray, and a two-sided chain. For every finite $r \geq 2$,*

$$|\mathcal{T}_r| = 2^{\aleph_0} = \mathfrak{c},$$

and the lower bound is already realized using permutations.

Proof. For one injection, each vertex has one outgoing edge and at most one incoming edge. A periodic component is precisely a finite cycle. A nonperiodic component either has a first vertex, giving a ray, or has a predecessor at every point, giving a two-sided chain. These are all the possibilities.

There are at most continuum many finite or countable codes in any fixed finite language. For the lower bound, for each $S \subseteq \omega$ define two permutations of $\mathbb{Z}$. The first is $F(n) = n + 1$. The second, G_S , swaps 0 with 1, swaps

$$3(m+1) \text{ with } 3(m+1)+1 \quad (m \in S), \quad (4.2)$$

and fixes all other integers. The transpositions are disjoint. An isomorphism respecting F must be a translation of $\mathbb{Z}$. The least integer moved by G_S is 0 for every S , so that translation must fix 0 and hence be the identity. It then respects G_S, G_T exactly when $S = T$. This gives continuum many different types. For $r > 2$, add identity permutations. $\square$

4.3 Definability without assuming the scheme being tested

Lemma 4.2 (Finite paths exist internally). *Reachability by a specified finite word in the operations f_i and their partial inverses is uniformly first-order definable in $\mathcal{L}_r$. Every successful finite path has an internal graph. These facts require no expanded Replacement.*

Proof. A forward letter i requires $J_i(a) = b$; a backward letter requires $J_i(b) = a$. Injectivity makes each backward step unique when it exists. A finite sequence of vertices witnessing the word is coded as an ordinary set function on a finite ordinal. It uses only finitely many atoms and therefore belongs to $M_\kappa(A)$, even when $\kappa = \omega$. Quantification over such finite graphs gives one first-order formula with the word as a pure parameter. The natural numbers here are the well-founded external natural numbers, not a nonstandard model's pseudo-finite sequence lengths. $\square$

Lemma 4.3 (Uniform recognition of component types). *If $\kappa > \omega$, there is a formula $\text{Type}_r(t, a)$ in $\mathcal{L}_r$ such that, for every $t \in \mathcal{T}_r$ and atom a ,*

$$\mathcal{N}_{\kappa, f} \models \text{Type}_r(t, a) \iff a \in B_t.$$

The same conclusion for finite component types holds for $\kappa = \omega$.

Proof. Ask for a code $c \in t$ and a set function h from its pure domain into the atoms, injective and with a in its range. Require h to preserve every labeled function. Require also that its range is closed under incoming edges: for each i ,

$$\forall b (\text{At}(b) \wedge J_i(b) \in \text{ran}(h) \longrightarrow b \in \text{ran}(h)).$$

Forward closure already follows from preservation of the code functions. The code is connected, so its image is connected; closure in both directions and inclusion of a make that image exactly the component of a .

Conversely, enumerate the actual component by a finite ordinal or by ω . Its code is pure. Its enumeration graph has only countably many atoms and is therefore internal when $\kappa > \omega$; for a finite component it is internal at every cutoff. The required witnesses exist. All conditions are first-order assertions about the set code and the set graph h . No appeal has been made to expanded Replacement or to a satisfaction predicate for arbitrary formulas. $\square$

Definition 4.4 (The small-type part). The small-type part of the atom graph is

$$D_\kappa(f) = \bigcup \{B_t : t \in \mathcal{T}_r, |B_t| < \kappa\}. \quad (4.3)$$

The adjective “small” applies to the *whole block of all components of one type*, not to an individual component. Every component is countable, but its type block may be arbitrarily large.

This distinction is decisive. Infinitely many copies of one finite cycle can form a large homogeneous block, while one copy of each of infinitely many different finite cycles forms a union of small, individually distinguishable blocks.

5 The exact Replacement criterion

Theorem 5.1 (Component-profile criterion). *Let $r \geq 1$ be finite and let f be an r -tuple of injections on A .*

1. *If $\kappa > \omega$, then*

$$\mathcal{N}_{\kappa,f} \models \text{Replacement}(J) \iff |D_\kappa(f)| < \kappa. \quad (5.1)$$

2. *If $\kappa = \omega$, then*

$$\mathcal{N}_{\omega,f} \models \text{Replacement}(J) \iff \begin{cases} \text{every component of } (A; f) \text{ is finite,} \\ D_\omega(f) \text{ is finite.} \end{cases} \quad (5.2)$$

5.1 Necessity: collecting uniquely definable type blocks

Proof of necessity for $\kappa > \omega$. Use the internal pure domain $\mathcal{T}_r$. In the expanded language define $F(t)$ as follows: if there is a set whose members are exactly the atoms satisfying $\text{Type}_r(t, a)$, let $F(t)$ be that set; if there is no such set, let $F(t) = \emptyset$. Extensionality for sets makes this a total functional definition. By [lemma 4.3](#), its external interpretation is

$$F(t) = \begin{cases} B_t, & |B_t| < \kappa, \\ \emptyset, & |B_t| \geq \kappa. \end{cases}$$

Indeed, a set of atoms belongs to $M_\kappa(A)$ precisely when its cardinality is less than κ .

Expanded Replacement yields an internal range set for F on $\mathcal{T}_r$. Taking its union produces exactly $D_\kappa(f)$. The kernel of a set of atoms is the set itself, so this union must have cardinality less than κ . $\square$

Proof of necessity for $\kappa = \omega$. First let a be an atom. The pure set of finite words in the $2r$ forward and backward letters is countable and belongs to the model. By lemma 4.2, a word either has a unique endpoint starting at a , or has no endpoint because a backward step is unavailable. Define its value to be that endpoint when it exists and a otherwise. This is a total functional definition in the expanded language. Its range is exactly the component of a .

Expanded Replacement would therefore make every component an internal set of atoms, hence finite. Once this is known, repeat the preceding type-block argument over the pure set $\mathcal{T}_r^{\text{fin}}$ of finite component types. The finite-type recognition formula is correct at the finite-kernel cutoff, and its union of small blocks is $D_\omega(f)$. That union must be finite. $\square$

The first argument uses formulas substantially more complicated than $y = J_i(x)$. Amenability alone says nothing about collecting all the type blocks selected by that formula.

5.2 Sufficiency: the fresh-component argument

For $P \subseteq A$, let $\text{cl}_f(P)$ be the union of all components meeting P . It is closed under all the f_i and all their available inverse steps. If $\kappa > \omega$ and $|P| < \kappa$, then

$$|\text{cl}_f(P)| \leq \max(\aleph_0, |P|) < \kappa. \quad (5.3)$$

For $\kappa = \omega$ the corresponding closure is finite whenever P is finite and all components are finite.

Lemma 5.2 (Fresh component swap). *Let $H \subseteq A$ be a union of components with $|H| < \kappa$. Let $y \in M_\kappa(A)$, and suppose $b \in \ker(y) \setminus H$ belongs to a type block B_t of cardinality at least κ . If $\kappa > \omega$, there is an atom permutation π commuting with every f_i , fixing H pointwise, and satisfying $\hat{\pi}(y) \neq y$. The same holds for $\kappa = \omega$ when the components are finite.*

Proof. The component C of b is disjoint from H . For uncountable κ , B_t contains at least κ distinct components: fewer than κ countable sets cannot have union of cardinality at least κ , since their sizes are uniformly bounded by $\aleph_0$. The set $H \cup \ker(y)$ meets fewer than κ components. Choose a component C' of type t disjoint from that set.

Choose a labeled-graph isomorphism $u : C \rightarrow C'$. Define π to be u on C , u^{-1} on C' , and the identity elsewhere. It commutes with each f_i because components are invariant in both directions and u respects the functions. It fixes H and moves b to an atom outside $\ker(y)$, so it changes that kernel and hence changes y .

At $\kappa = \omega$, a fixed finite type has a fixed finite component size. An infinite block of that type has infinitely many components, while $H \cup \ker(y)$ meets only finitely many. The same construction applies. $\square$

Proof of sufficiency in theorem 5.1. Assume the right side of the appropriate criterion. Fix a formula $\varphi(x, y, p)$ of $\mathcal{L}_r$ that defines a unique output for every x in an internal set w . Set

$$P = \ker(w) \cup \ker(p), \quad H = D_\kappa(f) \cup \text{cl}_f(P). \quad (5.4)$$

By the hypothesis and (5.3), H has size less than κ . At the finite cutoff this follows instead from finiteness of both terms. It is a union of components.

We claim that every uniquely defined output y has kernel contained in H . Otherwise choose $b \in \ker(y) \setminus H$. Its type block cannot be small, because every small block is included in $D_\kappa(f)$. Apply lemma 5.2. The resulting lifted automorphism preserves the *expanded* structure, fixes x and p , and changes y . It contradicts uniqueness.

External Replacement now collects all these unique outputs into a set Y . Its kernel is contained in H , independently of the cardinality of the input set w . Thus $Y \in M_\kappa(A)$, as required. The argument works for every fixed expanded-language formula and every internal parameter. $\square$

Remark 5.3 (Why the external construction is legitimate). The ambient parameters A, κ, f are sets, the model is definable, and each j_{f_i} has an external recursive definition. A fixed expanded formula therefore has an ordinary external first-order translation. External Separation and Replacement can be applied to it. The proof then checks whether the resulting set meets the internal kernel cutoff. It does not assume any expanded Replacement inside $\mathcal{N}_{\kappa, f}$.

5.3 Boundary examples for one symbol

Example 5.4 (A named automorphism can destroy Replacement). Let $\kappa = \aleph_\omega$ and choose a subset of A partitioned as $\coprod_{n < \omega} B_n$, with $|B_n| = \aleph_n$. On B_n , let f consist of cycles of length $n + 2$. Fix all remaining atoms. Every B_n is an internal set and is precisely the block of atoms of that cycle length. Their union has cardinality $\aleph_\omega$. Thus $D_\kappa(f)$ is not small and $\text{Replacement}(J)$ fails, although j_f is an amenable automorphism fixing every pure set.

A direct failed instance is the unique definition sending n to the set of all atoms of exact period $n + 2$. An internal range set for this definition would contain, in its kernel, the whole union of the B_n .

Example 5.5 (The two finite-cutoff obstructions are independent). At $\kappa = \omega$, a shift on a two-sided infinite atom chain has no small nonempty type block, but Replacement fails by collecting the iterates of one atom. Conversely, take one finite cycle of each length $n + 2$ on a countable subset of atoms and fix any remainder. Every component is finite, but the union of the finite cycle-type blocks is infinite, so Replacement fails by collecting those blocks. Both clauses in (5.2) are necessary.

Example 5.6 (Infinitely many atoms need not cause a failure). An arbitrary number of copies of one finite cycle type, with an infinite total block, is harmless at the finite cutoff provided the union of all the other finite blocks is finite. The main theorem applies because a unique output cannot select a new component from the homogeneous infinite block. The issue is distinguishability, not simply the number of atoms moved.

[write] **The finite-cutoff criterion for group actions, 4 October 2026, batch 90.** Batch-90 manuscript 05, *Named Symmetries and Replacement*, pinned to commit 8dc2592e9 (an ancestor of this report's placement, so it does not cite this report), is printed as Part VIII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (its source 12, labels hset:ns:). It works at the finite cutoff $M_\omega(A)$ with a uniformly named action of an arbitrary pure set-sized group Γ on the atoms. Its Theorem 58.1 (hset:ns:thm:full): expanded Replacement holds iff every Γ -orbit is finite and the union of the finite orbit types that occur only finitely often is finite. For finitely many named permutations this is part (2) of theorem 5.1: naming the atom permutations and naming their canonical lifts give definitionally equivalent languages (its Propositions 62.1 and 62.4), the orbits of the generated free-group action are the components here, and its exceptional set is $D_\omega(f)$ once

every component is finite. The proofs are independent; this report also covers non-surjective injections and every uncountable cutoff, which that part does not. New there: arbitrary pure group actions; the graded levels R_k (Replacement for outputs with at most k atoms), which hold for $k \geq 1$ iff every orbit is finite and only finitely many atoms have orbit of size at most k under the centralizer of the named action (its Theorem 59.1); one cycle of each length, as in [example 5.5](#), satisfies every R_k while failing Replacement (its Theorem 60.1 and Example 60.2); and, for every $d \geq 1$, two involutions, each harmless alone and with all joint orbits finite, whose joint expansion satisfies exactly the levels R_k with $k < d$ (its Theorem 61.3), a finite-cutoff, graded counterpart of [corollary 7.9](#). It also proves that pure-valued Collection, hence the whole pure sign-sequence surreal universe, survives every such naming (its Theorem 63.1). Its Remark 54.4 (hset:ns:rem:nee) records the overlap from its side. AI-assisted, unrefereed, not formalized; neither manuscript is claimed first.

6 Sharp thresholds for one name and for two names

Define $\text{Rob}_r(\kappa, A)$ to mean:

For every external r -tuple of injections on A , its canonical expanded model satisfies full Replacement in the r -symbol language.

This is an external property of a definable family of models. Its meaning is the same if “injections” is replaced by “permutations” in the equivalence below. The negative examples use permutations, and the positive arguments cover injections.

Theorem 6.1 (Finite-signature robustness thresholds). *For every infinite κ , every $|A| \geq \kappa$, and every fixed finite $r \geq 1$,*

$$\text{Rob}_r(\kappa, A) \iff \begin{cases} \text{cf}(\kappa) > \omega, & r = 1, \\ \text{cf}(\kappa) > \mathfrak{c}, & r \geq 2, \end{cases} \quad \mathfrak{c} = 2^{\aleph_0}. \quad (6.1)$$

6.1 The positive bounds

Proof of the forward preservation bounds. If $\text{cf}(\kappa) > \omega$, a countable union of sets of cardinality less than κ still has cardinality less than κ . There are only countably many one-injection types. Thus $D_\kappa(f)$ is small for every f , and [theorem 5.1](#) proves Rob_1 .

More generally, for $r \geq 2$ there are at most $\mathfrak{c}$ component types. If $\text{cf}(\kappa) > \mathfrak{c}$, the union of all the small blocks has cardinality less than κ . Again apply [theorem 5.1](#). These hypotheses automatically put us at an uncountable cutoff. $\square$

6.2 Sharpness for one symbol

Failure when $\text{cf}(\kappa) = \omega$. At $\kappa = \omega$, use a two-sided infinite shift on a countable subset of A , fixing the rest. Its infinite component violates [\(5.2\)](#).

For uncountable κ of countable cofinality, choose infinite cardinals $\lambda_n < \kappa$ cofinal in κ . On disjoint atom blocks B_n of sizes λ_n , use cycles of distinct lengths $n + 2$, and fix the remaining atoms. The union of these small type blocks has size κ . The argument of [example 5.4](#) applies without any special property of the aleph sequence. This proves the converse for $r = 1$. $\square$

6.3 Sharpness for two symbols: a real-coded construction

Lemma 6.2 (Cardinal decomposition). *If $\kappa > \omega$ and $\delta = \text{cf}(\kappa)$, there are infinite cardinals $\lambda_i < \kappa$ for $i < \delta$ with*

$$\sum_{i < \delta} \lambda_i = \kappa.$$

Proof. For regular κ , take $\delta = \kappa$ and every $\lambda_i = \aleph_0$. For singular κ , take an increasing cofinal sequence of infinite cardinals below κ . Its cardinal sum is κ . $\square$

Assume now $\kappa > \omega$ and $\delta = \text{cf}(\kappa) \leq \mathfrak{c}$. Choose pairwise distinct pure real codes $S_i \subseteq \omega$, $i < \delta$, and cardinals as in [lemma 6.2](#). On a subset A_0 of A of cardinality κ , use the atom indexing

$$A_0 = \{a_{i,\xi,n} : i < \delta, \xi < \lambda_i, n \in \mathbb{Z}\}. \quad (6.2)$$

The subscripts are external labels, not internal atom coordinates. Define permutations f, g by

$$f(a_{i,\xi,n}) = a_{i,\xi,n+1}, \quad g(a_{i,\xi,n}) = a_{i,\xi,G_{S_i}(n)},$$

where G_S is the involution in [\(4.2\)](#). On $A \setminus A_0$, let both maps be the identity.

Each set $\{a_{i,\xi,n} : n \in \mathbb{Z}\}$ is one component. Different i give nonisomorphic components by [lemma 4.1](#). For each i , the entire type block

$$B_i = \{a_{i,\xi,n} : \xi < \lambda_i, n \in \mathbb{Z}\}$$

has cardinality $\lambda_i < \kappa$. Their union is A_0 , of size κ . Hence $D_\kappa(f, g)$ is not small, and expanded Replacement fails.

For completeness, the failure can be expressed directly without the general quotient of component codes. On any constructed component, an atom a is the *root* exactly when $g(a) \neq a$ and every point strictly before a in the f -chain is fixed by g . This is uniformly first-order expressible using finite positive and negative iterates, as in [lemma 4.2](#). The root is its coordinate-0 atom. Its real label is recovered from

$$m \in S \iff g(f^{3(m+1)}(a)) \neq f^{3(m+1)}(a). \quad (6.3)$$

Thus, with the pure parameter $q = \langle S_i : i < \delta \rangle$, there is a single formula defining B_i as the set of atoms whose f -chain has a root with label S_i . Every B_i exists internally and is unique. Replacement for $i \mapsto B_i$ on the pure set δ would collect all of A_0 into a kernel of size κ , which is impossible.

At $\kappa = \omega$ a shift already defeats one symbol, and adding identity symbols cannot repair its failed instance. For $r > 2$, pad the two-symbol construction with identities. This completes the converse of [theorem 6.1](#).

Corollary 6.3 (A continuum-hypothesis characterization). *For any external atom set A of cardinality at least $\aleph_2$,*

$$\text{CH} \iff \text{Rob}_2(\aleph_2, A).$$

Equivalently, CH holds exactly when every pair of atom permutations has a joint canonical expansion of $M_{\aleph_2}(A)$ satisfying Replacement.

Proof. By [theorem 6.1](#), the preservation property is equivalent to $\aleph_2 > \mathfrak{c}$, since $\aleph_2$ is regular. By choice and Cantor's theorem, this is equivalent to $\mathfrak{c} = \aleph_1$. $\square$

This statement neither proves nor refutes CH. It translates CH into a precise preservation property for a family of language expansions. In particular, changing the ambient continuum can change a universal expansion property even though every one-symbol expansion at the same successor cutoff is harmless.

Corollary 6.4 (Countably many named maps). *The universal preservation threshold for a countable sequence of atom injections, with one symbol for each lift and ordinary finitary first-order formulas, is also $\text{cf}(\kappa) > \mathfrak{c}$.*

Proof. Each formula uses only finitely many symbols. Under the stated inequality its Replacement instance is valid by the finite-symbol theorem. Conversely the two-symbol counterexamples embed in a countable expansion by adding identities. $\square$

7 Collection, reflection, and their exact spectra

7.1 Collection on small domains

For an external cardinal μ , let $\text{Collection}_{\leq \mu}(J)$ denote all Collection instances on internal sets of cardinality at most μ , with arbitrary internal parameters. These cardinalities agree with their external values by [theorem 2.4](#).

Lemma 7.1 (Automatic small-domain Collection). *For every finite tuple f , if $\mu < \text{cf}(\kappa)$ then*

$$\mathcal{N}_{\kappa, f} \models \text{Collection}_{\leq \mu}(J).$$

This conclusion does not assume expanded Replacement.

Proof. Fix a set w of size at most μ and a total relation on it. In the ambient universe, first bound ranks of witnesses for this fixed formula and then choose a witness y_x for each $x \in w$. Equivalently, use ambient Collection followed by ambient choice. The union of at most $\mu < \text{cf}(\kappa)$ kernels, each of size less than κ , has size less than κ . Hence the external witness set $\{y_x : x \in w\}$ is internal and supplies Collection. $\square$

Lemma 7.2 (The limit-cutoff obstruction is already bounded). *Suppose κ is an infinite limit cardinal, including ω . There is a Collection instance with a Δ_0 matrix in the base language, on a pure domain of size $\text{cf}(\kappa)$, that fails in $M_\kappa(A)$ and in every expansion under consideration.*

Proof. Choose a pure set w of ordinals below κ , of size $\delta = \text{cf}(\kappa)$, whose cardinalities are unbounded below κ . For $\kappa = \omega$, take $w = \omega$. For uncountable limit κ , use a cofinal family of cardinals below κ .

Let $\text{InjectAtoms}(\alpha, s)$ assert that s is the graph of an injection with domain α and with atom values. It has a Δ_0 definition in $\{\in, \text{At}\}$; an explicit bounded formulation is given in [section A](#). For each $\alpha \in w$, an external injection into A has graph with $|\alpha| < \kappa$ atoms, hence exists internally.

If a single internal set v contained suitable graphs for all $\alpha \in w$, its kernel would have cardinality at least $|\alpha|$ for all those α . This contradicts $|\ker(v)| < \kappa$ and the unboundedness of their cardinalities. $\square$

Corollary 7.3 (Exact local spectrum at limit cutoffs). *For limit κ , every finite tuple f , and every cardinal μ ,*

$$\mathcal{N}_{\kappa, f} \models \text{Collection}_{\leq \mu}(J) \iff \mu < \text{cf}(\kappa).$$

Proof. Combine the preceding two lemmas. Failure on one domain of cardinality $\text{cf}(\kappa)$ implies failure for every larger domain bound. $\square$

7.2 A reservoir lemma at successor cutoffs

Lemma 7.4 (Compression into a fixed component reservoir). *Let $\kappa = v^+$ with v infinite, and suppose $|I(f)| \leq v$. For every component-closed $P \subseteq A$ of cardinality at most v , there is a component-closed $H \supseteq P$, $|H| \leq v$, such that every object $y \in M_\kappa(A)$ can be carried to an object supported on H by an expanded-structure automorphism fixing P pointwise.*

Proof. First add all small type blocks to P . There are at most v such blocks and each has size at most v , so their union is still of size at most v . Call the result P^* .

Every remaining nonempty block has size at least v^+ and therefore has at least v^+ components. For each such type choose v components disjoint from P^* . Let H consist of P^* and all these selected reservoirs. There are at most v types and all components are countable, so $|H| \leq v$.

The kernel of y meets at most v components of each type. Leave components in P^* fixed. For each large type, map the components meeting that kernel outside P^* injectively into the chosen v reservoir components. This partial matching extends to a permutation of the components outside P^* : its domain and image have size at most v , whereas the whole component family has cardinality at least v^+ . Choose compatible isomorphisms on matched components and arbitrary isomorphisms on the remainder. These component permutations together define an atom permutation commuting with all the f_i and fixing P^* . Its lift has the required properties. $\square$

Theorem 7.5 (Full Collection criterion). *For any infinite κ and finite nonempty tuple f ,*

$$\mathcal{N}_{\kappa,f} \models \text{Collection}(J) \iff \kappa \text{ is an uncountable successor cardinal and } |I(f)| < \kappa. \quad (7.1)$$

Proof. At limit cutoffs, including ω , lemma 7.2 rules out Collection. Suppose $\kappa = v^+$. If $|I(f)| \geq \kappa$, choose a pure subset $I_0 \subseteq I(f)$ of cardinality κ . By lemma 4.3, for every $t \in I_0$ there is an atom satisfying $\text{Type}_r(t, a)$. A witness set for all these types must contain at least κ atoms, since type blocks are disjoint. It cannot be internal. Thus Collection fails.

Conversely assume $|I(f)| < v^+$. Fix a total expanded relation $\varphi(x, y, p)$ on an internal set w . Close $\ker(w) \cup \ker(p)$ under components and apply lemma 7.4. For each $x \in w$, any witness can be moved by an automorphism fixing x, p to a witness with kernel contained in the same H . Externally choose such witnesses and bound their ranks, or use ambient Collection and choice. Their range set has kernel contained in H , hence is internal. This proves full expanded Collection. $\square$

Corollary 7.6 (All local Collection cases). *For a successor κ , if $|I(f)| < \kappa$ then Collection holds on every internal domain. If $|I(f)| \geq \kappa$, then*

$$\mathcal{N}_{\kappa,f} \models \text{Collection}_{\leq \mu}(J) \iff \mu < \kappa.$$

Together with corollary 7.3, this gives the exact domain-size spectrum in every case.

Proof. The positive statement is theorem 7.5. In the second case the preceding proof fails Collection on a domain of size κ , while lemma 7.1 proves it below $\text{cf}(\kappa) = \kappa$. $\square$

[write] **A second proof of the unexpanded spectrum, 3 October 2026, batch 89.** For the unexpanded model (one name, $f = \text{id}_A$, a single realized type) lemma 7.1, lemma 7.2, corollary 7.3 and theorem 7.5 give the Collection spectrum of $M_\kappa(A)$ that batch-89 manuscript 03 proves independently in Part VI of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (its delivered Lemma 10.1, Theorems 10.2–10.3 and Corollary 10.4; Lemma 33.1, Theorems 33.2–33.3 and Corollary 33.4 there; see Section 1.4). Both proofs stand. They differ in three places. (i) *The failing instance at a limit cutoff.* Manuscript 03 collects sets of atoms of prescribed cardinalities μ_i cofinal in κ , a relation that quantifies over bijections; lemma 7.2 collects injection graphs $\alpha \rightarrow A$ with the Δ_0 matrix of Appendix A, so the failure is already one of Δ_0 Collection. That bound, and proposition 8.1 built on it, are this report's own. (ii) *The successor case.* Manuscript 03 moves each witness, by an atom permutation fixing the parameters' kernels, into one fixed reservoir of ν atoms and bounds the stages by Replacement inside the model; lemma 7.4 must use permutations that commute with the named maps, so it moves whole components into a reservoir of components, one per realized type, and bounds ranks externally. For $f = \text{id}_A$ the two arguments coincide in substance. (iii) *What is named.* Manuscript 03's Theorem 11.2 (Theorem 34.2 there, `hset:zr:thm:expandedthreshold`) names an external enumeration $e : \kappa \rightarrow A_0$ of κ atoms by a predicate E ; then Replacement and Collection both hold below $\text{cf } \kappa$ and both fail on a domain of size $\text{cf } \kappa$, so at $\kappa = \aleph_1$ naming e destroys Replacement, whereas every one-name expansion by a canonical lift keeps full Collection and reflection there (the paragraph after theorem 7.7). There is no conflict: e is not an elementary lift, and every atom permutation preserving it fixes A_0 pointwise, so the fresh-component symmetries of Section 5 are unavailable. Manuscript 03's observation that its small-domain bound needs no invariance under atom permutations is the argument of lemma 7.1, which likewise holds in every expansion. Its Corollary 10.5 (Corollary 33.5 there; each infinite regular τ is the first failure size of some $M_{\aleph_\tau}$) is corollary 7.3 at $\kappa = \aleph_\tau$, a case this report does not state.

7.3 Full transitive reflection

In this section, full reflection in $\mathcal{L}_r$ has the following explicit meaning. For every finite set Φ of formulas and every internal set w , there is an internal transitive set t with $w \subseteq t$, closed under the named functions, such that each formula of Φ is absolute between the full expanded model and its restriction to t , for all its parameter tuples from t . Call this scheme $\text{RP}(J)$.

Theorem 7.7 (Exact reflection criterion). *In the present model family,*

$$\text{RP}(J) \iff \text{Collection}(J) \iff [\kappa \text{ is an uncountable successor and } |I(f)| < \kappa].$$

Proof. Full reflection implies Collection directly. Given a total relation on w , reflect a finite family containing its matrix and its totality assertion into a transitive set containing w, p . The restricted totality assertion supplies witnesses in that set, and absoluteness of the matrix makes them genuine witnesses in the full structure. Thus theorem 7.5 proves necessity.

For sufficiency, write $\kappa = \nu^+$ and assume $|I(f)| \leq \nu$. Enlarge the finite family Φ by subformulas, after expressing universal quantifiers through negation and existential quantifiers. We construct externally an increasing sequence

$$X_n = V_{\alpha_n}(H_n), \quad n < \omega,$$

where H_n is a component-closed atom set of size at most ν , the ordinals α_n increase, $w \subseteq X_0$, and every true existential subformula with parameters in X_n has a witness in X_{n+1} .

Begin by closing the atoms of w under components and choosing a sufficiently large α_0 . At stage n , apply lemma 7.4 to H_n . This gives a component-closed $H_{n+1} \supseteq H_n$, of size at most ν , into which witnesses can be compressed while all parameters in X_n are fixed. There are only set-many relevant

parameter tuples and finitely many subformulas. External Collection yields a common rank bound for witnesses after compression. Choose $\alpha_{n+1} > \alpha_n$ large enough that all these witnesses lie in $V_{\alpha_{n+1}}(H_{n+1})$.

Let $t = \bigcup_{n < \omega} X_n$. Its kernel is contained in $\bigcup_n H_n$, of size at most ν , so t is internal. It is transitive and closed under every J_i , since the H_n contain whole components and canonical lifts preserve rank. A finite parameter tuple in t lies in some X_n . The witness property and structural induction on the formulas now prove the required absoluteness. Atomic formulas agree because the restricted structure has the same membership, atom predicate, and function values; the Boolean steps are immediate; the existential step is exactly the witness property. $\square$

Remark 7.8 (Why one fixed atom reservoir is not a reflecting universe). A single $V_\alpha(H)$ can contain H as an element and consequently regard the atoms it sees as exhausted by one set. The full model has no set of all atoms. The reflection construction therefore grows the atom reservoirs along with the rank bounds. It does not incorrectly identify a fixed-reservoir hierarchy with a reflecting substructure of the entire atom universe.

For one named injection, at most countably many types are realized. Thus every one-symbol expansion at an uncountable successor cutoff satisfies full Collection and reflection. The unexpanded model is the special case $f = \text{id}_A$, and its full Collection/reflection criterion reduces to the familiar successor-versus-limit kernel cutoff.

7.4 Two individually safe names need not be jointly safe

Corollary 7.9 (Failure of compositional safety). *For $|A| \geq \aleph_1$, there are atom permutations f, g such that each of*

$$(M_{\aleph_1}(A), j_f), \quad (M_{\aleph_1}(A), j_g)$$

satisfies expanded Replacement, Collection, and full transitive reflection, but $(M_{\aleph_1}(A), j_f, j_g)$ fails expanded Replacement.

Proof. Use the construction (6.2) with $\delta = \aleph_1$ and every $\lambda_i = \aleph_0$. It is available because $\aleph_1 \leq \mathfrak{c}$. The joint expansion fails Replacement as proved in section 6. Each individual expansion is a one-symbol expansion at a successor cutoff, so theorems 7.5 and 7.7 apply; Replacement follows as well. $\square$

Example 7.10 (Joint Replacement can survive while joint Collection fails). Again put $\kappa = \aleph_1$ and use $\aleph_1$ distinct real labels. This time take $\aleph_1$ copies of the countable component for each label, on a subset of A of size $\aleph_1$. Each realized nontrivial type block has size $\aleph_1$. Any additional fixed-atom block is either large or is one small block. Hence $D_\kappa(f, g)$ has size less than κ , and joint Replacement holds. But there are $\aleph_1$ realized types, so Collection and full reflection fail. Both individual one-symbol expansions still have Collection and reflection.

Property	Controlling quantity	Exact criterion
Replacement, $\kappa > \omega$	Union of small type blocks	$ D_\kappa(f) < \kappa$
Replacement, $\kappa = \omega$	Component sizes and small blocks	All components finite and $D_\omega(f)$ finite
Full Collection	Cutoff and realized type count	κ successor and $ I(f) < \kappa$
Full reflection	Same as full Collection	Same criterion, with named-function-closed transitive reflectors
Universal one-name safety	Countably many possible types	$\text{cf}(\kappa) > \omega$
Universal finite multi-name safety	Continuum many possible types	$\text{cf}(\kappa) > \mathfrak{c}$ for at least two names

8 Partial reflection and a quantifier-normalization trap

Full reflection above requires agreement on a finite family for all parameters in the reflecting structure. A weaker question asks only that a single true assertion with its given parameters hold in some transitive set containing those parameters. In weak set theories it matters greatly which syntactic assertions are allowed.

Proposition 8.1 (Two superficially similar reflection schemes). *Over replacement-based set theory with atoms:*

1. Every true assertion of the form $\exists y \delta(y, p)$, where δ is Δ_0 in the base language, is true in some transitive set containing p .
2. The scheme of partial transitive reflection for assertions

$$\forall x \in w \exists y \delta(x, y, p), \quad (8.1)$$

with Δ_0 matrices δ , is equivalent to Δ_0 Collection.

Proof. For the first assertion, choose a witness y and take a transitive closure containing p, y . Bounded formulas are absolute between transitive sets and the universe: their quantifiers inspect only elements of already present parameters. Atomhood is unchanged. Hence the witness still works in the transitive set.

For the forward direction of the second assertion, reflect a true instance (8.1) into a transitive set t containing w, p . Every $x \in w$ belongs to t , and the reflected assertion supplies a witness $y \in t$. Bounded absoluteness makes that witness correct in the full universe. Thus t is a collection set.

Conversely, Δ_0 Collection supplies a set b of witnesses. A transitive closure containing w, p, b contains suitable witnesses for every $x \in w$. Bounded absoluteness makes (8.1) true in that closure. $\square$

At a limit cutoff, the instance from lemma 7.2 therefore has no such transitive reflector, even though each of its individual existential assertions does. This gives a precise warning about terminology. In the presence of Collection, bounded universal quantification in front of an existential

assertion can often be converted into one existential quantifier over a set of witnesses. Without Collection, that conversion is precisely what is at issue.

It is therefore unsafe to identify a strict existential–bounded normal form with every bounded-quantifier closure customarily placed at the same level of a hierarchy, unless the required normalization has been proved in the base theory. ProveIt’s bounded-consistency documentation explicitly separates its actual quantifier-group measure from the Levy hierarchy [5]; the present models give concrete semantic tests for such distinctions.

Remark 8.2 (No conflict with the Glazer–Yao separation results). The results of [section 7](#) concern a particular well-founded family with ambient choice, an explicit cutoff, and finitely many canonical function symbols. They do not reproduce the choiceless separation diagram in [1], and they do not assert that all reflection notions coincide over arbitrary urelement theories. Even within this paper, [proposition 8.1](#) distinguishes two restricted partial-reflection schemes that must not be silently conflated.

9 A concrete ProveIt formalization route

9.1 Inspected interfaces and the repository snapshot

The repository material used here is pinned to commit

2b7b388ba81a3355b19a7c2d2fe797e92f572355.

In particular, the inspected beginning of

SetTheory/ZF/Lean/ZF/Zf.lean

contains the formula constructors `Sep_form`, `Func_form`, `Image_form`, and `Repl_form`, and the axiom sets `ZFax` and `ZFax_s`. Its schema constructors act on the current formula type `Form`. Its introductory documentation describes the bridge from object-language axioms to semantic axioms and the internal finite-recursion development [4].

The bounded-consistency README states a numeralwise endpoint, one derivation for each externally fixed complexity bound, rather than an object-language universal consistency claim. It also records the choice of partial satisfaction rather than a reflection-based implementation [5]. We have inspected those interfaces and descriptions, not rebuilt or independently audited the repository’s claimed formal results.

The self-embedding report’s README explicitly marks its contents as unrefereed and not formalized [6]. That status matters: adjacency to verified Lean code does not convert a mathematical report into a kernel-checked theorem. The same boundary applies to this manuscript.

[write] Formal status at the write, 3 October 2026, batch 89. The interfaces described in this subsection are unchanged at the write (Section 1.4). None of the six layers below exists in ProveIt: the shared formula type `Form` of `Logic/FirstOrder/Lean/FirstOrder/Fol.lean`, used by `SetTheory/ZF` and `SetTheory/BoundedConsistency`, has only the atomic formulas $\in$ and $=$, with no atom predicate and no function symbols, and no urelement theory, kernel model or canonical lift is formalized. The module names and interfaces of this section are proposals.

9.2 First task: make the language parameter explicit

A port should not merely append an “embedding axiom” to the existing pure-membership theory. It needs a genuine expanded syntax. Two reasonable routes are available.

The direct route introduces a language with a unary atom predicate and r unary function symbols. Terms, renaming, substitution, satisfaction, and all formula-valued schema arguments must then be extended. This is mathematically transparent but requires term infrastructure if the selected existing syntax is purely relational.

A relational route instead uses binary graph predicates $E_i(x, y)$, together with totality and functionality axioms. It can be closer to the repository’s current relational syntax. Formulas containing function terms can be translated to graph formulas, but the translation may introduce quantifiers. Therefore an equivalence of theories is not automatically a bound-preserving equivalence of proof-complexity measures.

For either route, there must be separate semantic assertions

$$\text{ModelsReplacement}(\mathcal{L}_U, M) \quad \text{and} \quad \text{ModelsReplacement}(\mathcal{L}_r, M, J).$$

The second quantifies over a larger formula type. The counterexamples of [section 6](#) should become regression tests against any interface that mistakenly identifies these assertions.

9.3 A proof dependency plan

A practical implementation can be divided into six layers.

Layer 1: well-founded objects with atoms. Implement a tagged cumulative model, or a well-founded extensional relation with distinguished atoms, and define its kernel. Prove kernel arithmetic, the hereditary cutoff property, and the existence of all pure codes used later. Keep the ambient cardinal assumptions visible.

Layer 2: recursive lifts and semantic automorphisms. Define the lift of an atom injection by well-founded recursion. Prove its kernel equation and functoriality. Prove the small-partial-bijection extension lemma, then base-language elementarity by local agreement with automorphisms. Do not derive this by assuming the lift is surjective.

Layer 3: component combinatorics. Define finite labeled paths, undirected reachability, and component closure. Prove the countable component bound from finite degree, and prove the component-swap lemma with an explicit isomorphism witness. This layer is independent of the set-theoretic Replacement scheme.

Layer 4: internal recognition of types. Define a set of pure graph codes and its isomorphism quotient. Prove the satisfaction specification for the type-recognition formula. The cardinal hypothesis $\omega < \kappa$ must be attached exactly where a countable enumeration graph is imported into the model. At the finite cutoff, use finite codes only after proving that expanded Replacement forces components to be finite.

Layer 5: schemes and spectrum theorems. Represent the type-block selector as one expanded formula and prove that Replacement for it imports the union of small blocks. Prove sufficiency using the kernel bound (5.4). Then formalize the marker construction, the cardinal decompositions, and the universal thresholds. Collection needs its own relation and reservoir arguments; it should not be inferred from Replacement.

Layer 6: reflection and proof-theoretic interfaces. Implement reflection only for an externally fixed finite set of formulas, using the growing-reservoir construction. Relate semantic instances to the chosen object theory through explicitly proved soundness or completeness bridges. No uniform truth predicate or internal assertion of full elementarity is needed.

9.4 Expected hard points and trust boundaries

The finite graph swap is elementary; the demanding work is likely to be the internal syntax and the exact specifications of set-coded component isomorphisms. There are four particularly important failure modes to test.

First, a port must not use expanded Replacement to construct the finite paths or countable enumeration graphs later used to *test* Replacement. Their membership in the cutoff model is proved externally from their kernels. Second, a cardinal-kernel bound is a bound on the atoms in a whole set of outputs, not on each output independently. Third, class elementarity must remain a scheme unless the metatheory explicitly supplies an appropriate satisfaction class. Fourth, the reflecting sets must be closed under every named function, not merely under membership.

The supplementary Python program checks finite component swaps and the real-marker coding on finite test families. It does not prove any infinite-cardinal theorem, verify a first-order derivation, or replace the proposed Lean development. Its purpose is to catch combinatorial implementation mistakes before a formal port.

10 Further research questions

The following are proposed projects arising from the present results. They are not all claimed to be previously published open problems. Their formulation records where the proved classification ends.

Question 10.1 (An intrinsic definable-closure theorem). For a fixed finite tuple f , determine the exact atom kernel of the definable closure of a parameter set in the expanded model. The proof of [theorem 5.1](#) bounds it by component closure plus $D_\kappa(f)$, but this bound can be larger than necessary. Which atoms or entire finite blocks are actually definable, rather than merely protected from a fresh-component swap?

[write] Answered for commuting permutations, 4 October 2026, batch 93. For a finite tuple of *commuting permutations* the bound is exact: [theorem 17.1](#) of Part II proves, at every infinite cutoff including $\kappa = \omega$, that the atom kernel of the objects uniquely definable from $\bar{a}$ is exactly $G \cdot \bigcup_i \ker(a_i) \cup D_\kappa(\bar{p})$, the orbit saturation (here the component closure) of the parameter kernels together with the whole small-type part, and that pure parameters add nothing. [Example 17.4](#) shows that this formula fails as stated for a non-surjective injection at $\kappa = \omega$ (the root of a ray is definable). The question stays open for injections, for non-commuting tuples, and for the atom-valued part of definable closure; Questions [23.1](#) and [23.2](#) of Part II continue it.

Question 10.2 (Noninitial elementary embeddings). Classify elementary self-embeddings of $M_\kappa(A)$ without requiring transitive range. Can one obtain a useful dichotomy separating canonical atom lifts from embeddings that alter the pure core or introduce noncanonical members into images of sets? Any such theorem must state its external class-existence assumptions explicitly.

Question 10.3 (Fixed-parameter rigidity beyond canonical lifts). For canonical lifts, a proper parameter-definable map cannot fix a parameter defining it. Investigate analogous statements for noninitial maps, for embeddings between different cutoffs, and for richer predicates on atoms. The key issue is whether the stabilizer argument still forces the atom action to be small.

Question 10.4 (Commuting named automorphisms). The continuum-sized type family used in the two-name lower bound is noncommutative. Determine the sharp threshold for a fixed number of *commuting* atom permutations. Their components are transitive actions of a finitely generated abelian group, whose stabilizer structure is much more restricted. This suggests that commutation can lower the type-count threshold, but the uniform internal recognition and all boundary cases should be proved separately.

[write] **Answered, 4 October 2026, batch 93.** [Theorem 15.3](#) of Part II answers this question in full: for every fixed $r \geq 1$, every infinite κ and every atom set with $|A| \geq \kappa$, every r -tuple of commuting atom permutations keeps expanded Replacement if and only if $\text{cf}(\kappa) > \omega$, the threshold of a single name. The boundary cases are covered: at $\kappa = \omega$ a shift fails, and at uncountable cutoffs of countable cofinality the distinct-cycle construction of the proof of [theorem 6.1](#) for $r = 1$ ([example 5.4](#) at $\aleph_\omega$), padded by identities, fails. The uniform internal recognition the question asks for is [lemma 15.2](#) (stabilizer classes are parameter-free definable at every cutoff); the type count is [lemma 15.1](#). Part II goes further: commuting permutations together with t commuting injections have threshold $\text{cf}(\kappa) > \omega$ for $t \leq 1$ and $\text{cf}(\kappa) > \epsilon$ for $t \geq 2$ ([theorem 16.3](#)); for a finitely generated group named by generators the threshold is $\text{cf}(\kappa) > \tau(G)$, the number of conjugacy classes of subgroups, at uncountable κ , and finiteness of G at $\kappa = \omega$ ([theorem 18.2](#)). For finitely many commuting involutions, and more generally a finite group, at $\kappa = \omega$ the answer was already in the repository as Corollary 58.2 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:cor:finitigroup`; see the note after [corollary 15.4](#)).

Question 10.5 (Other bounded-degree signatures). Extend the component-profile criterion to relational structures on atoms of uniformly bounded or countable degree. Identify hypotheses ensuring that component types have pure set codes recognizable internally without the Replacement scheme under examination. The fresh-component sufficiency argument is flexible; necessity depends on the definability of the complete type blocks.

Question 10.6 (Uncountably many names and infinitary syntax). With ordinary finitary formulas, even arbitrarily many names are tested one finite sublanguage at a time. What changes for a specified infinitary logic that admits countable conjunctions or longer formulas? Component sizes, available type codes, satisfaction specifications, and cofinality bounds should all be recalibrated together.

Question 10.7 (General atom ideals). Replace $[A]^{<\kappa}$ by a downward-closed, finite-union-closed atom ideal. Find an exact analogue of [theorem 5.1](#) in terms of ideal-small unions of isomorphism blocks and ideal-small component closures. Cardinal symmetry was used both to extend partial matchings and to find fresh components; those two requirements may separate for a general ideal.

Question 10.8 (Removing ambient choice). Which parts of the criteria survive without ambient choice? Enumeration of countable components, extension of partial bijections, and the conversion from a bound on witnesses to an internal set each require separate analysis. This is the most direct route toward the choiceless reflection phenomena studied by Glazer and Yao, rather than merely staying in a choice-based special case.

Question 10.9 (Minimal complexity of failed instances). Determine the least syntactic complexity of a formula witnessing failure of expanded Replacement in the real-coded two-name construction. The limit-cutoff Collection obstruction is already Δ_0 , but the component-block selector uses unbounded quantification. Can the two-name obstruction be lowered to a sharp prenex level without importing Collection through normalization?

Question 10.10 (Forcing and preservation properties). The CH characterization suggests a systematic forcing analysis of $\text{Rob}_r(\kappa, A)$. Track which cutoffs, atom reservoirs, and external class parameters are held fixed when the pure continuum changes. Distinguish preservation of one already given expanded structure from universal preservation over all tuples in a forcing extension.

[write] **Fixed-action part answered, 4 October 2026, batch 93.** [Theorem 21.1](#) of Part II answers the fixed-action part: for a ground tuple of injections on a ground atom set, set forcing that keeps κ a cardinal does not change whether the rebuilt expansion satisfies expanded Replacement (component types and the small-type part $D_\kappa(f)$ are absolute). It does not assert that the old class model or the truth value of each formula is unchanged, nor that Collection is absolute under collapses below κ . Open: what happens when κ is collapsed and the model is rebuilt at the new cutoff, and the universal properties $\text{Rob}_r(\kappa, A)$, whose counterexamples may be new actions of the extension ([Question 23.11](#) of Part II).

Question 10.11 (Bounded consistency with atoms and named maps). Can ProveIt's externally indexed partial-satisfaction construction be adapted to a replacement-based urelement theory without accidentally adding Collection? After that, determine exactly which enlarged-language schema instances are needed to formalize the component-profile proof. A theorem about an external family of bounded-consistency derivations must remain distinct from its universal closure in the object language.

Question 10.12 (Machine-checked independence certificates). Develop a reusable Lean interface that packages a definable class model, its valid schemes, and one explicit failed instance of another scheme. Use the one-name singular-cutoff example and the two-name successor-cutoff example as test cases. A good interface should make it impossible to transfer Replacement across a language expansion without providing a proof of the corresponding new instances.

[write] **Status of the questions, 3 October 2026, batch 89.** None of the twelve questions is answered in ProveIt. The two that name the repository remain open there: SetTheory/BoundedConsistency proves its numeralwise endpoint for pure ZFC only, with no atoms and no named maps, and no Lean interface of the kind the last question asks for exists. On the question about removing ambient choice: Part VII of Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets (batch-89 manuscript 04) works in choiceless permutation models with atoms, but about surreal coding and universality; it says nothing about named lifts, Replacement or Collection in the models here. [Added 4 October 2026, batch 90:] Part VIII of the same surreal report (batch-90 manuscript 05) asks the finite-cutoff counterparts of four of these questions for named group actions: its Question 65.1 is the question on general atom ideals, its Question 65.3 the one on other bounded-degree signatures, its Question 65.6 the one on minimal complexity of failed instances, and its Question 65.12 the one on machine-checked certificates; for the question on commuting named automorphisms, its Question 65.4 notes that finitely many commuting involutions generate a finite group and so never destroy Replacement at the finite cutoff (its Corollary 58.2). None of these is answered there. Conversely, [theorem 5.1\(1\)](#) and [theorem 6.1](#) answer in part its Question 65.2 (infinite kernel bounds), and [theorems 7.5](#) and [7.7](#) its Question 65.11 (reflection and Collection in expanded models), for finitely many named injections; dated notes after those two questions there record this. See also the note at the end of [section 5](#). [Added 4 October 2026, batch 93: the first sentence of this note is no longer true. Part II of this report (batch-93 manuscript 01) answers [Question 10.4](#) in full, [Question 10.1](#) for commuting permutations and the fixed-action part of [Question 10.10](#); dated notes after those three questions give the details. Questions 10.1 and 10.10 stay open beyond those parts, and the other nine questions are

untouched. Part II's own questions (Section 23) continue Questions 10.1, 10.6, 10.7, 10.8, 10.9, 10.10 and 10.12, as the note after them records.]

11 Conclusion and proof-status summary

The central invariant is not the number of atoms moved by an elementary map, nor the existence of all its set-sized restrictions. It is the union of the small, internally recognizable component-type blocks of the atom structure obtained after the maps are named.

For uncountable cutoffs, that union is small exactly when full expanded Replacement holds. At the finite cutoff, finiteness of all components is an additional necessary and sufficient condition. Counting the available component types gives the sharp distinction between one name and two: the thresholds are ω and $2^{\aleph_0}$ in the cofinality comparison. Counting the *realized* types, together with the successor/limit nature of the cutoff, gives the exact Collection and full-reflection criterion.

These statements are proved for an explicitly constructed family over an external set of atoms. The foundational construction is known; the proposed new content is the expansion analysis and its sharp classifications. The article does not settle CH, does not contradict results about elementary embeddings of pure universes, and does not claim that an external semantic argument has already been turned into a Lean proof.

The most concrete next mathematical tests are the commuting-automorphism case and the minimal complexity of the two-name failure. The most concrete implementation test is whether a formal library can distinguish base-language Replacement, amenability of named maps, and Replacement over the expanded formula type throughout the entire development.

[write] **Added 4 October 2026, batch 93.** The commuting-automorphism case is settled in Part II (theorem 15.3); the minimal complexity of the two-name failure remains open (Question 23.10).

A An explicitly bounded injection-to-atoms formula

Here is a formulation supporting the Δ_0 claim in lemma 7.2. All bounded quantifiers below are ordinary abbreviations for quantification restricted by membership. No function symbols for pairing or union are required.

Define the bounded predicates

$$\begin{aligned} \text{Sing}(u, a) &\equiv \text{Set}(u) \wedge a \in u \wedge \forall z \in u (z = a), \\ \text{Double}(v, a, b) &\equiv \text{Set}(v) \wedge a \in v \wedge b \in v \wedge \forall z \in v (z = a \vee z = b), \\ \text{OP}(p, a, b) &\equiv \text{Set}(p) \wedge \exists u \in p \exists v \in p (\text{Sing}(u, a) \wedge \text{Double}(v, a, b) \\ &\quad \wedge \forall z \in p (z = u \vee z = v)). \end{aligned}$$

Thus $\text{OP}(p, a, b)$ says that $p = \langle a, b \rangle$ in Kuratowski coding.

The formula $\text{InjectAtoms}(\alpha, s)$ is the conjunction of $\text{Set}(s)$ and the following three bounded conditions.

Every graph entry has the right shape.

$$\forall p \in s \exists i \in \alpha \exists q \in p \exists a \in q (\text{At}(a) \wedge \text{OP}(p, i, a)).$$

Every domain element is represented.

$$\forall i \in \alpha \exists p \in s \exists q \in p \exists a \in q (\text{At}(a) \wedge \text{OP}(p, i, a)).$$

Functionality and injectivity. For all $p, p' \in s, q \in p, q' \in p', a \in q, a' \in q',$ and $i, i' \in \alpha,$ require

$$(\text{OP}(p, i, a) \wedge \text{OP}(p', i', a')) \longrightarrow (i = i' \leftrightarrow a = a').$$

Every displayed quantifier is bounded. The first clause excludes spurious graph entries; the second gives totality on α ; the last simultaneously imposes single-valuedness and injectivity. For $\alpha = 0$, the clauses force $s = \emptyset$. Because the domain elements used in the application are pure ordinals, atom values cannot be confused with those indices.

Transitive absoluteness of this formula is immediate from structural induction on bounded formulas. In particular, a transitive set containing an alleged injection graph sees its actual entries and their atom values, rather than some truncated graph with hidden range atoms.

B A dependency audit of the principal proofs

Result	Positive mechanism	Point that must not be assumed
Base Replacement	Fresh atom transposition fixes inputs and moves a proposed new output atom.	Arbitrary witnesses need not be uniquely definable.
Elementary lift	On each finite parameter tuple, the atom injection agrees with an atom permutation.	The global injection need not extend to a global permutation agreeing everywhere.
Parameter-definability	Stabilizers of parameter kernels force identity off the kernel; internal graph codes give the converse.	External definability from the full atom map is not internal parameter-definability.
Type recognition	A countable component enumeration graph lies below an uncountable kernel cutoff.	Expanded Replacement is not used to construct that graph.
Necessity in the profile theorem	Replacement applied to one uniformly defined selector collects all small type blocks.	The blocks need not already form an internal family.
Sufficiency in the profile theorem	A fresh isomorphic component yields an automorphism commuting with every named map.	Separate commutation with only one of several names is insufficient.
Two-name lower bound	Marker roots and their real labels distinguish continuum many countable component types.	No continuum hypothesis is used to obtain $\aleph_1$ distinct reals.

Result	Positive mechanism	Point that must not be assumed
Successor Collection	A common reservoir of at most ν components per type bounds all witness kernels.	A rank bound alone does not bound the atoms in the witnesses.
Full reflection	Grow both rank bounds and component reservoirs for a fixed finite subformula family.	One fixed atom reservoir need not reflect the absence of a set of all atoms.

C Reproducibility and supplementary finite checks

The source package is self-contained: `article.tex` contains its bibliography and compiles with pdfLaTeX. The build script runs three passes to settle the table of contents and cross-references. No external mathematical software or bibliography processor is needed for the manuscript.

The supplementary `verify_finite.py` uses only Python’s standard library. It enumerates all permutations on at most seven points for the one-map swap checks, all ordered pairs of permutations on at most five points for the two-map checks, and all marker codes using six bits. It constructs component isomorphisms explicitly, verifies that the induced swaps commute with all named maps, and checks the finite-support marker involutions and their translation-rigidity property.

These are exact finite regression tests of the combinatorial constructions, not statistical experiments and not a formal verification of the theorems. They cannot establish a claim about an infinite cutoff, Replacement, a proper-class elementary embedding, or CH. The accompanying JSON file records the executed checks; the source audit records the repository pin and the literature used.

[write] Shipped layout and replay, 3 October 2026, batch 89. In ProveIt the program is `code/verify_finite.py`, the build script `code/build.sh`, and the recorded run and the delivered build audit are `data/verification_results.json` and `data/BUILD_AUDIT.json`; the source audit is `SOURCE_AUDIT.md` beside this article. The delivered `build.sh` builds the `article.tex` of its own directory, so in this layout it must be run on a copy with the article beside it. At the write the program was rerun on a scratch copy (Python 3.14.4): it passed, and its report equals the recorded one except for the field `python_version` (3.14.4 against the delivered 3.13.5). The delivered `BUILD_AUDIT.json` describes the delivered 27-page PDF and lists the hashes of the delivered files, among them the delivered PDF and README, which are not shipped; the article and PDF in ProveIt are the written versions. The report’s README gives the replay commands.

Part II

Commuting actions and Replacement

Exact cofinality thresholds, definable kernels, and uniform naming

A solution to ProveIt Question 10.4 with further classification theorems

Research manuscript prepared with ChatGPT, 4 October 2026

Status and scope. This article supplies complete arguments for its stated results in an explicitly specified family of models. The atom-kernel construction and the component-profile criterion are prior work, credited and reproved where needed. The commuting, mixed-injection, definable-kernel, and uniform-naming refinements are the contributions developed here. The work answers a question recorded as open in the inspected ProveIt report; publication priority beyond that report is not established. These are mathematical proofs, not machine-checked proofs. Elliot Glazer and Bokai Yao are cited researchers, not authors or endorsers of this manuscript.

Abstract. Let A be an external set of atoms, let κ be an infinite cardinal with $|A| \geq \kappa$, and retain exactly those sets whose transitive closures involve fewer than κ atoms. Naming an atom injection by its canonical elementary lift can change which instances of Replacement hold. We determine exactly what commutation contributes to this phenomenon. Every fixed finite nonempty family of commuting atom permutations preserves expanded Replacement for every action precisely when $\text{cf}(\kappa) > \omega$. More generally, for commuting signatures with s designated permutations and t additional injections, the threshold is ω when $t \leq 1$ and is $2^{\aleph_0}$ when $t \geq 2$. The latter bound is witnessed by explicit lattice staircases; the two coordinate maps can individually preserve Replacement and have only pure fixed points while their joint expansion fails Replacement.

For a finitely generated group G , the uncountable-cutoff threshold is the number of conjugacy classes of subgroups of G , with a separate and exact finite-cutoff criterion. For commuting permutations we also determine the complete atom kernel of definable closure, including the finite-cutoff case. Finally, an action of $\bigoplus_{\omega} C_2$ on two-point orbits shows that naming every group element separately can preserve Replacement at every cutoff while a single amenable binary evaluation map for the same action destroys it. The article explains the relation to Glazer–Yao reflection theory, gives explicit failed schema instances, distinguishes established results from the new refinements, and proposes a focused research program.

[write] Part II of this report since 4 October 2026: batch-93 manuscript 01 of ProveIt’s incoming-reports intake, printed in full (label prefix `nee:ca:`). It continues Part I and answers its Question 10.4 ([Question 10.4](#)), its Question 10.1 for commuting permutations ([Question 10.1](#)) and the fixed-action part of its Question 10.10 ([Question 10.10](#)). Section n of the manuscript is Section $n + 11$ here, so its item $n.k$ is item $(n + 11).k$ (its Theorem 4.3 is [theorem 15.3](#)); its Appendices A and B are Appendices D and E; its equations, numbered consecutively there, are numbered within sections here. “ProveIt Question 10.4” and every other number of Part I that the manuscript quotes are unchanged. Provenance, the overlap with Part VIII of birthday-cutoffs-and-hereditary-sets (prior in the repository), the imported group-theoretic fact and the notation are in Sections [12.4](#) and [12.5](#).

12 The research intersection and the results

The subject is the interaction between symmetry and axiom schemes in set theory with atoms. An atom is an object which may belong to sets but has no members of its own. Once atoms are

permitted, a universe may satisfy Replacement and the Axiom of Choice while lacking a set that contains all its atoms. This makes it possible to examine exactly how extra named structure changes definability.

Glazer and Yao's *Reflection Principles in ZFU* develops the interaction of Collection, reflection, and choice in urelement set theory, including constructions controlled by small atom kernels [1]. The directly relevant ProvelIt report, *Naming Elementary Embeddings*, studies canonical lifts of atom maps in a particularly explicit kernel model [7]. Its Question 10.4 asks for the sharp Replacement threshold when finitely many named atom permutations commute. The same report's Question 10.1 asks how sharp its bound on definable kernels can be.

We answer Question 10.4 for every positive finite number of names. We also give an exact answer to the kernel question in the commuting-permutation setting and extend the analysis to commuting injections and uniformly named group actions. The setting is deliberately concrete: the conclusions below concern the specified models, rather than all models of urelement set theory or all elementary embeddings.

12.1 The basic model and the meaning of preservation

The atom kernel $\ker(x)$ is the set of atoms occurring anywhere in the transitive closure of $\{x\}$. In an ambient universe with Choice, define

$$M_\kappa(A) = \{x \in U(A) : |\ker(x)| < \kappa\}. \quad (12.1)$$

Here $U(A)$ is the cumulative universe over the external *set* A of atoms. The model contains all pure sets, so it contains arbitrarily large pure sets even when only finitely many atoms may occur in any one of its objects.

An injection $f : A \rightarrow A$ extends recursively to a map j_f on sets. The base model regards j_f as an external elementary map; adjoining a symbol J for it creates new first-order formulas. *Expanded Replacement* means that Replacement is valid for every formula of this enlarged language, with arbitrary internal set parameters. It is stronger than the assertion that the graph of j_f restricted to each internal set is internal.

Throughout the article $\mathfrak{c} = 2^{\aleph_0}$ is computed in the ambient pure universe. A statement of *universal preservation* quantifies over all actions of the indicated kind on the fixed atom set A . Failure of this universal statement means that a counterexample action exists; it does not mean that every action fails.

12.2 Main classification in one table

The following table summarizes the universal statements. Every row assumes κ infinite and $|A| \geq \kappa$.

Named structure	Exact condition for universal Replacement
A fixed finite nonempty family of commuting permutations	$\text{cf}(\kappa) > \omega$
Commuting permutations together with at most one additional injection, with at least one name	$\text{cf}(\kappa) > \omega$
At least two additional commuting injections	$\text{cf}(\kappa) > \mathfrak{c}$
An action of a fixed finite group, named by generators	Every infinite κ
An action of a finitely generated G , with $\kappa > \omega$	$\text{cf}(\kappa) > \tau(G)$
An action of a finitely generated G , with $\kappa = \omega$	G is finite

Here $\tau(G) = |\text{Sub}(G)/G|$, where G acts on its subgroups by conjugation. In the mixed rows, an “additional injection” is not required to be surjective; it is permitted to be a permutation. The lower bounds can be made genuinely nonsurjective where stated below.

The first row answers the repository question. The second and third identify a finer boundary: one noninvertible direction is still controlled by countably many types, whereas two such directions admit a binary sequence of boundary data. The group formulation explains why a number of generators alone cannot determine the answer.

12.3 What is inherited and what is proved here

Result or ingredient	Provenance and role
The model $M_\kappa(A)$ and base Replacement	A specialization of the kernel-ideal constructions in urelement set theory; see Yao [3] and Glazer–Yao [1]. We give the needed symmetry proof.
The component-profile criterion	Theorem 5.1 of the inspected ProvelT report [7]. It is stated and reproved in section 14 , and is not claimed as a new result.
Transitive group actions as coset actions	Standard group theory [9]. The new use is the exact Replacement classification by the resulting type count.
Commuting and mixed signatures	Theorems proved here, including the boundary cases requested by ProvelT Question 10.4.
Definable atom kernels	An exact refinement proved here for commuting permutations, addressing Question 10.1 in that setting.
Separate names versus uniform evaluation	An explicit separation proved here. It uses ordinary finitary first-order syntax on both sides.
Collection and reflection	The underlying criteria are already in [7]; we explain their commuting-action consequences without claiming a new general reflection theorem.
Fixed-action forcing stability	A profile-based refinement proved here, answering the fixed-action part of Question 10.10 when the cutoff remains a cardinal.

The repository source was inspected at commit a9ab9a69872c2fb7464097af32fd7b76c75cb030.

The source’s dated status note says that its twelve proposed questions remain unanswered there. This is a precise, limited novelty claim: the present work resolves the commuting question recorded in that source. No claim is made to have settled an open problem expressly posed in Glazer and Yao’s paper, or to have established historical priority for every corollary.

12.4 Provenance and place in this report

[write] Added 4 October 2026, batch 93. *Source.* This Part is a single manuscript, printed in full: manuscript 01 of batch 93 of ProveIt’s incoming-reports intake, delivered as the archive `commuting_actions_replacement.zip` (690,120 bytes; inner directory `commuting_actions_replacement/`, main file `commuting_actions_replacement.tex`, 1,283 lines, 61 labels, 8 bibliography entries, a 31-page US Letter PDF) in the arrival commit `2faa3b37a`, and placed in commit `47a77daba` as Part II of this report (files prefixed `02-commuting-actions-`, labels `nee:ca:`). It was filed here rather than beside Part VIII of `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` because it is written as a continuation of Part I and answers Part I’s questions. The author line reads “Research manuscript prepared with ChatGPT”, 4 October 2026; the Part is AI-assisted, unrefereed and not formalized. Being a single source, it needed no merge choices: every statement, proof, example, remark, question and non-claim of the manuscript is printed. The write prefixed all 61 labels with `nee:ca:` and updated every reference; gave the twelve questions of Section 23 labels; replaced the manuscript’s title page and abstract environment by the Part heading, the status box and the paragraph “Abstract”; dropped its separate table of contents; set its questions on the shared theorem counter (their numbers do not change, since Section 23 contains nothing else numbered); numbered its equations within sections (in the manuscript they run consecutively: its (1) is (12.1), (2)–(3) are (13.1)–(13.2), (4)–(5) are (14.1)–(14.2), (6) is (15.1), (7)–(11) are (16.1)–(16.5), (12) is (17.1), (13)–(14) are (18.1)–(18.2) and (15)–(17) are (19.1)–(19.3)); pointed its figure at the shipped file; merged its bibliography into the report’s (below); and added these dated **[write]** notes, the notation table of Section 12.5 and Question 23.13. Item $n.k$ of the manuscript is item $(n + 1).k$ here; its Appendices A and B are Appendices D and E. No statement, proof or non-claim was changed.

Pin and repository claims. The pin `a9ab9a69872c2fb7464097af32fd7b76c75cb030` (Section 12 above, Appendix E and the bibliography entries [7, 8]) is the ProveIt commit “Audit eight arithmetic compiler reports and the complete degree-12 matrix DAG” of 4 October 2026, an ancestor of the placement. The manuscript’s quoted blob `5426fc8da5f470efc448c3dff041751d0d7dc4a9` is exactly this report’s `article.tex` at the pin; it differs from Part I as printed only by the two dated notes of the batch-90 reciprocal commit `e50dde15b`, so every number of [7] that the manuscript quotes (Theorems 5.1, 6.1, 7.5, 7.7; Questions 10.1, 10.4, 10.10) is the number in Part I: [theorems 5.1, 6.1, 7.5 and 7.7](#) and [Questions 10.1, 10.4 and 10.10](#). The manuscript’s other repository statements are true at the pin and at this write: `SetTheory/ZF/Lean/ZF/Zf.lean` (blob `b33db57cf`, the same at Part I’s pin `2b7b388ba`, at this pin and at the write) declares `Sep_form`, `Func_form`, `Image_form`, `Rep1_form`, `ZFax` and `ZFprov`, and no urelement model is formalized; the dated note of Part I said that its twelve questions were unanswered, which was true until this Part. No repository statement is contradicted, and nothing needed retraction.

Formal status. No statement of this Part is formalized in Lean or Rocq; the formal-status note of Section 1.4 applies unchanged (the formula type of `SetTheory/ZF` has no atom predicate and no function symbols), and the milestones of Section 22 are proposals.

Prior in the repository: Part VIII of the surreal report. Batch-90 manuscript 05, *Named Symmetries and Replacement*, is Part VIII of `birthday-cutoffs-and-hereditary-sets` (its source 12, labels `hset:ns:`). It works only at the finite cutoff, with a uniformly named action of an arbitrary pure set-sized group. Its text was written in commit `1404038df` (4 October 2026, 10:57), twelve minutes after this manuscript’s pin (10:45; the batch-90 placement `12076b2e8`, an ancestor of the pin, staged only its files), so this manuscript does not cite it. Several results here meet it at $\kappa = \omega$; Part VIII is prior in the repository, the proofs are independent, and both stand. At $\kappa = \omega$: [corollary 15.4](#) is its Corollary 58.2 (`hset:ns:cor:finitigroup`); the finite-cutoff clause of [theorem 18.2](#) follows from its Theorem 58.1 (`hset:ns:thm:full`) with its Proposition 62.1 (`hset:ns:prop:finitewords`); [proposition 19.1](#) is its Proposition 62.2 (`hset:ns:prop:language`) with

Corollary 58.2; [proposition 19.2](#) is Theorem 58.1, which there holds for every pure set-sized group; [proposition 19.4](#) is Proposition 62.1, there for free-group words; and [theorem 19.3](#) makes the separation of its Theorem 62.3 (`hset : ns : thm : evaluator`) with the same group $\oplus_{\omega} C_2$ but a different witness (below). New here: everything at uncountable cutoffs ([theorem 18.2](#) for $\kappa > \omega$, [proposition 19.2](#) for $\kappa > \omega$, [corollary 19.5](#)), and the witness of [theorem 19.3](#), whose orbits have at most two atoms, which works at every κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, whose evaluator is amenable, and which has a converse for $\text{cf}(\kappa) > \mathfrak{c}$; Part VIII’s witness is the regular action, with infinite orbits, at $\kappa = \omega$ only. Dated notes after each of these results record the overlap, and a note after [Question 23.12](#) relates the questions. The rest of the Part (commuting and mixed signatures, staircases, definable kernels, the forcing theorem) has no counterpart there.

Relation to Part I. The manuscript reproves and credits Part I’s foundations: [lemma 13.2](#) is [lemma 2.3](#), [proposition 13.3](#) is [theorem 2.4](#) (Yao’s construction, specialized; also Theorem 32.2 of Part VI of birthday-cutoffs-and-hereditary-sets, `hset : zr : thm : supportbase`), [proposition 13.4](#) is part of [theorem 3.2](#), [lemma 14.1](#) is [lemma 4.3](#) with [lemma 4.2](#), and [theorem 14.2](#) is [theorem 5.1](#). [Lemma 14.4](#) abstracts the sharpness constructions of [theorem 6.1](#) and [lemma 6.2](#); [theorem 16.4](#) is a commuting, proper-injection variant of [corollary 7.9](#), whose two permutations do not commute; [theorem 20.1](#) is a consequence of [theorems 7.5](#) and [7.7](#), as the manuscript says, and its failure at limit cutoffs is [lemma 7.2](#).

An imported fact. [Remark 18.4](#) rests on an existence theorem of Ivanov and Ol’shanskii [12], read in Smith’s paper [11]; it is not proved here. The note after that remark records what was checked, and [Question 23.13](#) records what is missing. Nothing else in this Part is stated without proof, and no claim of the manuscript was found to be wrong.

The bibliography. The manuscript’s entries for [1] and [3] are Part I’s and are printed once (the manuscript adds the full-text address <https://arxiv.org/html/2602.21970v1> and dates Yao’s dissertation “2023” with arXiv v3 “June 2023”). Its entry RepoZF, pinned at a9ab9a698, is cited as [8], so that Part I’s entry [4], pinned at 2b7b388ba, stays as it was; the file is the same blob at both pins. The entries [7, 9, 10, 11, 12] are new.

Non-claims of the delivery README (not shipped), beyond those in the text: the manuscript is not peer reviewed, and the “independent review” within its preparation (Appendix E) is not external review; it claims the resolution of the repository’s question only, not worldwide priority, and not the settlement of an open problem posed by Glazer and Yao, whose work is a credited source, not an endorsement or coauthorship; the scripts supplement the proofs and establish no infinite classification, cardinal arithmetic, forcing theorem, class-model theorem or axiom scheme; and the upstream repository was inspected, not modified.

Checks at the write. The delivered finite checks were rerun on a scratch copy (Python 3.14.4, standard library, about 2 s): they pass, with 347,278 explicit checks, and the output is byte-identical to the recorded one. The delivered figure generator, run on a scratch copy with Matplotlib 3.10.8, regenerates a byte-identical `staircases.pdf` (TrueType fonts, no Type 3) and a pixel-identical `staircases.png` (RGBA, 1970×1659 ; differently compressed). Section 22 and Appendix E name the delivered files; in Provel they are `code/02-commuting-actions-verify_examples.py`, `code/02-commuting-actions-make_figure.py`, `data/02-commuting-actions-verification_output.txt` and `figures/02-commuting-actions-staircases.pdf` (with its PNG preview). The generator writes `figures/staircases.*` beside itself, so it must be run on a copy; the report’s README gives the commands. Appendix E says that a LinkedIn page identifies Glazer; it gives no address, and none is recorded here.

12.5 Notation of Part II

[write] Added 4 October 2026, batch 93. No symbol was renamed; the only changes are the label prefix and one citation key (above). The objects of Part I keep their meaning: $M_{\kappa}(A)$ is $M_{\kappa}(A)$ of (2.1), and $\ker, j_f, \mathcal{N}_{\kappa,f}, \mathcal{T}_r, B_t, D_{\kappa}(f), V_{\alpha}(C), \mathfrak{c}$, Replacement, Collection, RP are as there. Several letters carry more than one meaning in this Part:

Symbol	Meaning here (and what it is not)
$\mathbb{N}$	the natural numbers. <i>Not</i> Part I's macro $\mathcal{N}_{\kappa,f}$, the expanded model
U_S	the staircase (16.4) coded by $S \subseteq \mathbb{N}$; <i>not</i> the cumulative universe $U(A)$
F, G	in Section 16, the two coordinate maps of a staircase; elsewhere G is a group: $\mathbb{Z}^r$ in Section 17, a finitely generated group in Section 18, a countable group or $\bigoplus_{\omega} C_2$ in Section 19. In lemma 4.1 of Part I, G_S is an involution of $\mathbb{Z}$
H	in Section 16, the group $\mathbb{Z}^S$ acting through the permutations; in Section 17, $H(\bar{a})$ is the protected union $G \cdot P \cup D_{\kappa}(p)$, the counterpart of (5.4); in Sections 18–19, a subgroup; in Section 20, a reservoir of atoms, as in lemma 7.4. Never the hereditary-size universe H_{κ}
K	a subgroup (a stabilizer) in Sections 15–17 and proposition 19.1; in the proof of theorem 16.4, a set of atoms
p, p_i, p	in Sections 16–17, named commuting permutations; elsewhere p is a parameter tuple, as in Part I
B_K	the atoms with stabilizer exactly K : the type block B_i of $t = \mathbb{Z}^r / K$
$E(g, x)$	the uniform evaluator (19.1) of a group action. It is Part VIII's evaluator $T(i, a, b)$ (hset:ns:thm:evaluator) and its named action graph Act (source 12's E) of birthday-cutoffs-and-hereditary-sets; <i>not</i> the enumeration predicate E of Part VI there, which names an injection of κ into the atoms (see the note after corollary 7.6)
$\tau(G), \tau_{\text{fin}}(G)$	numbers of conjugacy classes of all subgroups, of finite-index subgroups; <i>not</i> Part VIII's involution τ nor Part VI's $\tau = \text{cf } \kappa$ there
$\mathcal{K}(\text{dcl}(\bar{a}))$	scheme notation for the kernels of uniquely definable objects (before theorem 17.1); no truth predicate
universal preservation	preservation for every action of the stated kind on A : for r arbitrary injections this is Part I's $\text{Rob}_r(\kappa, A)$

13 Foundations, kernels, and canonical lifts

13.1 External and internal sets

Work in a well-founded ambient set theory with the Axiom of Choice and an external set A of urelements. Equivalently, ordinary ZFC can implement the entire construction with tagged atoms, tagged sets, and an interpreted membership relation. The use of $U(A)$ as a cumulative class does not require an inaccessible cardinal. The cardinal bounds used below are external cardinal bounds.

Write $\text{At}(x)$ for “ x is an atom” and $\text{Set}(x)$ for its negation. Extensionality applies to sets. For every object x put

$$\ker(x) = \{a \in A : a \in \text{TC}(\{x\})\}.$$

Thus $\ker(a) = \{a\}$ for an atom, whereas a pure set has empty kernel. A set of atoms $B \subseteq A$ belongs to $M_{\kappa}(A)$ exactly when $|B| < \kappa$.

The distinction between “ B is an external set” and “ B is an internal set of $M_{\kappa}(A)$ ” is essential. Every atom is internal. The totality A is an external set but is not an internal set, because $|A| \geq \kappa$. Conversely, every pure set is internal, with no restriction on its cardinality or rank.

For an atom set C , use the hierarchy

$$V_0(C) = C, \quad V_{\alpha+1}(C) = \mathcal{P}(V_{\alpha}(C)) \cup C, \quad V_{\lambda}(C) = \bigcup_{\alpha < \lambda} V_{\alpha}(C).$$

Every object supported on C occurs in some $V_\alpha(C)$. Each displayed stage is an external set. When $|C| < \kappa$, every object in every stage is internal.

Remark 13.1 (The semantic convention). All assertions about a formula holding in $M_\kappa(A)$ mean the external relativization of that *fixed* formula to the definable class (12.1). Elementarity, Replacement, reflection, and definable-closure assertions are interpreted as schemes. We do not posit an internal truth predicate for a proper-class structure or an internal set of all class embeddings. Whenever a proof collects witnesses, it uses external Collection for the fixed translated formula and then checks the kernel bound. This convention also governs the definable-kernel theorem in section 17.

13.2 Why the base model satisfies Replacement

The following elementary identities organize the argument:

$$\begin{aligned} y \in x &\implies \ker(y) \subseteq \ker(x), \\ \ker(\{x, y\}) &= \ker(x) \cup \ker(y), \\ \ker(\mathcal{P}(x)) &\subseteq \ker(x), \\ \ker(Y) &= \bigcup_{y \in Y} \ker(y) \quad \text{for an external set } Y. \end{aligned} \tag{13.1}$$

A finite union of sets of cardinality below an infinite κ still has cardinality below κ , even if κ is singular. It follows that $M_\kappa(A)$ is transitive, closed under subsets of its internal sets, and closed under Pairing, Union, and Powerset.

Every permutation π of A lifts recursively to an automorphism $\hat{\pi}$ of $U(A)$ and of $M_\kappa(A)$. If π fixes $\ker(x)$ pointwise, then $\hat{\pi}(x) = x$.

Lemma 13.2 (Base-language confinement). *If a formula of $\{\in, \text{At}\}$ uniquely defines y from internal parameters x, p , then*

$$\ker(y) \subseteq \ker(x) \cup \ker(p).$$

Proof. Suppose an atom b occurs in y but not in x or p . Since the union of their three kernels has size below κ , there is $c \in A$ outside that union. Interchanging b and c fixes x, p and changes the kernel of y . Its lift is a base-language automorphism and therefore produces a second output for the same uniquely defining formula. This is impossible. $\square$

Proposition 13.3 (The base theory). *The model $M_\kappa(A)$ satisfies the usual urelement axioms with Separation, Replacement, and the assertion that every set can be well-ordered. Its pure part is the full ambient pure universe. Its internal cardinality for each internal set agrees with the external cardinality.*

Proof. The structural axioms follow from (13.1). For a functional formula on an internal domain w , external Replacement supplies its range Y . Applying lemma 13.2 separately to each input gives

$$\ker(Y) \subseteq \ker(w) \cup \ker(p),$$

so Y is internal. Separation holds because every externally separated subset of an internal set has no new atoms.

An external well-order of w has kernel contained in $\ker(w)$, and hence belongs to the model. The graph of any external bijection between w and its pure cardinal has the same kernel bound. All pure objects are present, so there are no additional restrictions on pure ordinals, cardinal arithmetic, or the pure coding sets used below. $\square$

Collection is not included in the conclusion. In urelement set theory with an internally proper class of atoms, it is a distinct issue; [section 20](#) states the relevant distinction explicitly.

13.3 Elementary and amenable lifts

For an injection $f : A \rightarrow A$, define

$$j_f(a) = f(a), \quad j_f(x) = \{j_f(z) : z \in x\} \quad \text{if } \text{Set}(x).$$

Then

$$\ker(j_f(x)) = f[\ker(x)]. \quad (13.2)$$

The recursive map preserves ranks and maps $M_\kappa(A)$ to itself. It fixes every pure object. Composition of atom maps corresponds to composition of their lifts; in particular commuting atom maps have commuting lifts.

Proposition 13.4 (Canonical lifts). *Every j_f is a base-language elementary self-embedding of $M_\kappa(A)$. For each internal set w , the restriction graph $j_f \upharpoonright w$ is internal. If f is a permutation, j_f is an automorphism.*

Proof. Let $\bar{x}$ be a finite tuple of parameters and put $P = \bigcup_i \ker(x_i)$. The partial bijection $f \upharpoonright P : P \rightarrow f[P]$ extends to a permutation π of A : both complements have cardinality $|A|$, because $|P| < \kappa \leq |A|$. On every object in the tuple, $\hat{\pi}$ agrees with j_f . An automorphism preserves the truth of any fixed base-language formula, which proves elementarity on $\bar{x}$.

The graph of $j_f \upharpoonright w$ has kernel contained in $\ker(w) \cup f[\ker(w)]$, a set of size below κ . This is the asserted amenability. An inverse atom permutation lifts to an inverse automorphism when f is bijective. $\square$

Definition 13.5 (The expanded model). For a finite tuple $f = (f_1, \dots, f_r)$ of atom injections, let

$$\mathcal{N}_{\kappa, f} = (M_\kappa(A), \in, \text{At}, j_{f_1}, \dots, j_{f_r}).$$

The symbols $J_1, \dots, J_r$ are genuine function symbols. Expanded Replacement permits them inside every formula of the scheme. A lifted atom permutation is an automorphism of this expanded structure whenever its atom action commutes with each f_i .

Expanded Separation always holds: an externally defined subset of an internal set remains internal. The issue for Replacement is that a set of individually small outputs may collectively involve too many atoms.

14 The component-profile criterion

This section supplies the central prior tool in a self-contained form. Its content is Theorem 5.1 of [7]. The later classifications depend on the number and definability of its component types.

14.1 Codes and entire type blocks

Make the atoms into a labelled directed graph, with an i -edge from a to $f_i(a)$. A component is a connected component of the underlying undirected graph. There are at most $2r$ neighbors at each vertex, since an injection gives at most one predecessor of each label. Every component is therefore finite or countably infinite.

Let $\mathcal{T}_r$ be the pure set of isomorphism types of connected structures with r total injections on a nonempty finite ordinal or on ω . A type may be represented by the set of all its codes, so no definable choice of representative is needed. Define

$$B_t = \{a \in A : \text{the component of } a \text{ has type } t\}, \quad D_\kappa(f) = \bigcup_{\substack{t \in \mathcal{T}_r \\ |B_t| < \kappa}} B_t. \quad (14.1)$$

The word “block” always refers to all copies of one component type. A countable component can belong to a block of arbitrarily large cardinality.

Lemma 14.1 (Recognition without expanded Replacement). *For $\kappa > \omega$, membership $a \in B_t$ is uniformly first-order definable in the expanded language, with the pure type t as a parameter. At $\kappa = \omega$ the same holds for finite component types. A specified finite path is definable at every cutoff.*

Proof. A path is represented by a function on a finite ordinal. A forward step of label i requires $J_i(a) = b$; a backward step requires $J_i(b) = a$. The graph of a successful finite path uses finitely many atoms, so it is internal even at $\kappa = \omega$.

For a full type, ask for a pure code $c \in t$ and an injective set function h from its domain into the atoms. Require a in its range, preservation of every labelled operation, and closure of its range under incoming edges. The image is connected, closed under both directions of every edge, and contains a , so it is exactly the component of a . Conversely an enumeration of the actual component has a graph with at most countably many atoms. It is internal when $\kappa > \omega$, and it is internal at $\kappa = \omega$ when the component is finite. These are external kernel checks on particular graphs, not applications of expanded Replacement. $\square$

Theorem 14.2 (Component-profile criterion; prior work). *For a nonempty finite tuple of atom injections,*

$$\mathcal{N}_{\kappa, f} \models \text{Replacement} \iff |D_\kappa(f)| < \kappa \quad (\kappa > \omega).$$

At $\kappa = \omega$, expanded Replacement holds exactly when every component is finite and $D_\omega(f)$ is finite.

Proof. Necessity at an uncountable cutoff. On the pure domain $\mathcal{T}_r$, define $F(t)$ to be the set of all atoms of type t if such an internal set exists, and $\emptyset$ otherwise. By lemma 14.1 this is a single first-order definition. Set Extensionality makes its output unique. Its external value is B_t exactly when $|B_t| < \kappa$, and is empty otherwise. Replacement would collect the range of F into an internal set. The union of that range is $D_\kappa(f)$, so this union must have size below κ .

The additional finite-cutoff obstruction. Fix an atom a . On the pure set of finite words in the forward and backward labels, define the value to be the endpoint of the path starting at a , using a as a default when a backward step does not exist. The endpoint is uniquely specified by a finite graph. Its range is the entire component of a . Expanded Replacement at $\kappa = \omega$ would make this component finite. Once every component is finite, the preceding selector argument over the pure finite types proves that D_ω is finite.

Sufficiency. For a set P of atoms let $\text{cl}(P)$ be the union of components meeting P . If $\kappa > \omega$ and $|P| < \kappa$, then

$$|\text{cl}(P)| \leq \max(\aleph_0, |P|) < \kappa. \quad (14.2)$$

At the finite cutoff the closure of a finite P is finite when every component is finite. Suppose a formula uniquely specifies y_x for each x in an internal domain w , with parameter tuple p . Put

$$H = \text{cl}(\ker(w) \cup \ker(p)) \cup D_\kappa(f).$$

The relevant right-hand conditions make H small. It is a union of components.

If $b \in \ker(y_x) \setminus H$, its type block has size at least κ . For an uncountable κ this block has at least κ components: each has at most $\aleph_0$ vertices, and a family of fewer than κ uniformly countable sets has union smaller than κ . Fewer than κ components meet $H \cup \ker(y_x)$, so there is a fresh component of the same type disjoint from that set. At $\kappa = \omega$, the same conclusion holds because this particular type has a fixed finite size and its block has infinitely many copies.

Interchange the component of b with the fresh component by a labelled isomorphism and its inverse, fixing every other component. This is an atom permutation commuting with every f_i . It fixes x, p , and it moves b outside the old kernel of y_x . Its lift changes the unique output, a contradiction. Therefore every output kernel is contained in H . External Replacement collects the outputs into a set with kernel contained in H , which is internal. This proves every expanded instance. $\square$

Remark 14.3 (Where singular cardinal arithmetic is used). The proof does not assert that an arbitrary union of fewer than κ small sets is small. That assertion can fail when κ is singular. In (14.2) there is a *uniform countable bound* on component sizes. Later unions of whole type blocks require the stronger comparison with $\text{cf}(\kappa)$.

Lemma 14.4 (From a supply of types to a sharp bound). *Suppose a class of finite-signature atom structures has at most θ countable connected types, is closed under disjoint unions, and admits a fixed harmless type for unused atoms. At an uncountable cutoff, $\text{cf}(\kappa) > \theta$ ensures universal Replacement. If θ is infinite and there are θ distinct types available independently, then every $\kappa > \omega$ with $\text{cf}(\kappa) \leq \theta$ admits a counterexample.*

Proof. For the positive direction, a union of fewer than $\text{cf}(\kappa)$ sets, each of size below κ , has size below κ . Apply this to the small type blocks.

For the converse put $\delta = \text{cf}(\kappa)$. Choose δ distinct nonreserved types t_i and infinite cardinals $\lambda_i < \kappa$ with $\sum_{i < \delta} \lambda_i = \kappa$. If κ is regular use $\delta = \kappa$ and $\lambda_i = \aleph_0$; if it is singular use a cofinal sequence of infinite cardinals below it. Give type t_i exactly λ_i copies. Its block has size λ_i because every component is nonempty and countable. Realize their disjoint union on κ atoms and use the reserved type on any remainder. The small blocks together contain κ atoms. The failed instance is the type-block selector in the proof of [theorem 14.2](#). $\square$

15 Commuting permutations: the exact answer

Fix $r \geq 1$. A tuple of r commuting permutations gives an action of the free abelian group $\mathbb{Z}^r$, where the standard basis vectors act by the named permutations. The action may have a kernel; faithfulness is not required.

15.1 Orbit classification and finite subgroup descriptions

Lemma 15.1. *Connected structures of r commuting permutations are exactly the transitive $\mathbb{Z}^r$ -sets $\mathbb{Z}^r / K$ for subgroups $K \leq \mathbb{Z}^r$. Two such labelled structures are isomorphic exactly when their subgroups are equal. There are countably infinitely many types.*

Proof. For an atom a , the orbit map $u \mapsto u \cdot a$ identifies its orbit with $\mathbb{Z}^r / \text{Stab}(a)$. In a transitive action stabilizers change by conjugation; for an abelian group they do not change at all. Hence a labelled isomorphism preserves K , and equal K gives the evident isomorphism.

Every subgroup of $\mathbb{Z}^r$ is finitely generated. Here is a proof that also records the finitary information we need. Induct on r . Project a subgroup $K \leq \mathbb{Z}^{r-1} \times \mathbb{Z}$ to its last coordinate. Its image is $d\mathbb{Z}$. If $d = 0$, use induction in $\mathbb{Z}^{r-1}$. If $d > 0$, choose $v \in K$ with last coordinate d . Subtracting a suitable integer multiple of v from any member of K leaves an element of $K \cap (\mathbb{Z}^{r-1} \times \{0\})$, which is finitely generated by induction. Those generators together with v generate K . Thus the set of subgroups is the image of a countable set of finite integer matrices. It is infinite, as witnessed by $n\mathbb{Z} \times \mathbb{Z}^{r-1}$ for positive n . $\square$

The proof classifies labelled actions, not merely the abstract quotient groups. Changing the basis of $\mathbb{Z}^r$ would change the interpretation of the names, and is not an allowed isomorphism here.

Lemma 15.2 (Stabilizers can be recognized at every cutoff). *For each fixed subgroup $K \leq \mathbb{Z}^r$, the class of atoms with stabilizer exactly K is parameter-free definable in the expanded language. The definition is correct even at $\kappa = \omega$.*

Proof. Fix a finite integer matrix generating K . Its entries can be built into a formula by fixed numerals, and membership in its integer span is a first-order assertion about pure integers. A vector $u \in \mathbb{Z}^r$ determines a finite word in the named permutations and their inverses. Its value on a is expressed by a finite path graph, which is internal by [lemma 14.1](#). The required predicate says

$$\text{At}(a) \wedge \forall u \in \mathbb{Z}^r (u \cdot a = a \iff u \in K). \quad (15.1)$$

This abbreviates a first-order formula using the finite matrix and finite-path definition. It quantifies over words individually; it does not first assemble an orbit or a simultaneous table of all their values. All pure integer coding is available in $M_\kappa(A)$. $\square$

Theorem 15.3 (Commuting-permutation threshold). *For every fixed $r \geq 1$, every infinite cardinal κ , and every atom set A with $|A| \geq \kappa$, the following are equivalent:*

- (i) *Every r -tuple of commuting atom permutations has a canonical expansion of $M_\kappa(A)$ satisfying expanded Replacement.*
- (ii) $\text{cf}(\kappa) > \omega$.

Proof. If $\text{cf}(\kappa) > \omega$, then $\kappa > \omega$. By [lemma 15.1](#) the small-type part is a countable union of sets of size below κ , so it is small. Apply [theorem 14.2](#).

If $\kappa = \omega$, let the first permutation shift a two-sided countably infinite orbit and let all other names act identically. Replacement on finite iterates of an atom would collect an infinite atom set, so it fails.

If $\kappa > \omega$ and $\text{cf}(\kappa) = \omega$, choose infinite cardinals $\lambda_n < \kappa$ cofinal in κ . On disjoint blocks B_n of size λ_n , let the first permutation consist of cycles of exact length $n + 2$, and make every other

permutation the identity. Fix all atoms outside these blocks. The B_n are precisely the entire blocks of those cycle types. Each is internal, but $|\bigcup_n B_n| = \kappa$. The formula sending n to the complete set of atoms of exact period $n + 2$ is total and single-valued and has no internal range. This is the required failed instance. $\square$

This proves the threshold and both boundary cases requested in ProveIt Question 10.4. An arbitrary fixed finite number of commuting permutations therefore has the countable type threshold of one permutation.

[write] **Question 10.4 answered, 4 October 2026, batch 93.** “ProveIt Question 10.4” is [Question 10.4](#) of Part I; a dated note there records the answer. The positive half (countably many types, [lemma 15.1](#)) is new; the two counterexamples are those of [theorem 6.1](#) for one name, padded by identity names.

Corollary 15.4 (Finite group relations). *If the named permutations generate an action of a fixed finite group, expanded Replacement holds at every infinite cutoff. In particular this is true of finitely many commuting permutations with fixed finite orders.*

Proof. A finite group has finitely many subgroups and all of its orbits are finite. Hence there are finitely many types and D_κ is a finite union of small blocks. Both clauses of the finite-cutoff criterion are satisfied as well. $\square$

Example 15.5 (Why a fixed order bound matters). One arbitrary permutation may have cycles of every finite length and destroy Replacement at $\aleph_\omega$. An involution has only fixed-point and two-cycle types, so it never does. A finite tuple of commuting involutions acts through a finite elementary abelian 2-group and has the same unconditional preservation property.

[write] **Prior in the repository at the finite cutoff, 4 October 2026, batch 93.** At $\kappa = \omega$, [corollary 15.4](#) and the example above are Corollary 58.2 of Part VIII of birthday-cutoffs-and-hereditary-sets (hset:ns:cor:finitigroup: every action of a finite pure group preserves Replacement, in particular one permutation of finite order), and that Part’s Question 65.4 (hset:ns:q:generators) notes that commuting involutions generate a finite group. Part VIII was written after this manuscript’s pin (Section 12.4); the proofs are independent. For $\kappa > \omega$ the corollary is immediate from [theorem 14.2](#), as the proof shows.

Remark 15.6 (Sharpness persists with faithful actions). For uncountable-cutoff counterexamples one can require the $\mathbb{Z}^r$ -action to be faithful. Reserve a countable regular $\mathbb{Z}^r$ -orbit, and use the cycle construction on the remaining κ atoms. Its stabilizer type differs from the finite-cycle types, so the bad small blocks are unchanged. At the finite cutoff a regular orbit already witnesses failure. Faithfulness therefore does not improve [theorem 15.3](#).

16 Commuting injections and the role of invertibility

Fix nonnegative integers s, t with $s + t > 0$. Consider pairwise commuting atom maps

$$p_1, \dots, p_s, f_1, \dots, f_t,$$

where the p_i must be permutations and the f_j need only be injections. Thus the maps give an action of the commutative monoid $\mathbb{Z}^s \times \mathbb{N}^t$ in which every acting map is injective. The classification depends on t , rather than on $s + t$.

16.1 One injection over a finitely generated abelian action

Let $H = \mathbb{Z}^s$ act through the p_i , and let f be one commuting injection. The case $s = 0$ is included by taking H trivial.

Lemma 16.1 (Orbit quotient and its normal forms). *A connected structure $(C; p_1, \dots, p_s, f)$ is determined, up to labelled isomorphism, by the following data:*

- (i) *a subgroup $K \leq H$;*
- (ii) *a quotient shape: a one-sided ray, a two-sided chain, or a cycle of some positive length n ;*
- (iii) *in the cycle case, an element $v \in H/K$.*

In particular there are countably many possible connected types.

Proof. Commutation makes f equivariant for the H -action. It sends the orbit $H \cdot a$ onto the orbit $H \cdot f(a)$. The induced map on the set of H -orbits is injective: if $f(b) = h \cdot f(a)$, then $f(b) = f(h \cdot a)$, so $b = h \cdot a$.

Moreover $\text{Stab}_H(a) = \text{Stab}_H(f(a))$. One inclusion follows from equivariance. If $h \cdot f(a) = f(a)$, injectivity of f gives $h \cdot a = a$, proving the other inclusion. Stabilizers are constant along H -orbits because H is abelian. Thus every fiber in a connected quotient component is a copy of the same $Q = H/K$.

The quotient graph of f is connected and has one outgoing edge and at most one incoming edge at each vertex. It is a finite cycle, a ray, or a two-sided chain. Along each edge, f gives an equivariant bijection between two copies of Q .

For a ray choose a point a_0 in its initial fiber and put $a_i = f^i(a_0)$. The map

$$Q \times \mathbb{N} \longrightarrow C, \quad (q, i) \longmapsto q \cdot a_i$$

identifies the structure with the normal form in which H translates the first coordinate and $f(q, i) = (q, i + 1)$. For a two-sided chain, choose a point in one fiber and use the unique inverse fiber maps to define a_i for negative i as well. The same formula gives $Q \times \mathbb{Z}$ with the shift in its second coordinate. These choices only prove existence of an isomorphism; no canonical internal choice is being asserted.

For a quotient cycle of length n , choose a_0 and set $a_i = f^i(a_0)$ for $0 \leq i < n$. There is a unique $v \in Q$ with $f^n(a_0) = v \cdot a_0$. In the resulting coordinates,

$$f(q, i) = \begin{cases} (q, i + 1), & i < n - 1, \\ (q + v, 0), & i = n - 1. \end{cases} \quad (16.1)$$

Changing the chosen point in a fiber does not change v , since translations commute. Moving the starting fiber around the cycle also preserves the same n -step translation. Conversely the displayed data construct the structure and distinguish its labelled isomorphism type.

There are countably many K by the finite-generation argument in [lemma 15.1](#), countably many possible n , and countably many cosets v for each K . This proves countability of the type set. There are infinitely many types already among the finite cycles with all p_i trivial. $\square$

The element v is the residual translation after one turn around the cycle, often called its holonomy. No such datum survives on a ray or a chain because the successive choices of fiber coordinates remove each edge translation without encountering a cycle.

16.2 Two noninvertible directions encode a real

We next build continuum many connected types of two commuting injections. A slightly stronger construction than is needed makes both individual maps look as simple as possible.

For $S \subseteq \mathbb{N}$, define an integer sequence d_S on $\mathbb{Z}$ by

$$d_S(m) = \begin{cases} 1, & m < 0, \\ 3, & m = 0, \\ 1 + \mathbf{1}_S(m-1), & m > 0. \end{cases} \quad (16.2)$$

Let $h_S(0) = 0$ and extend uniquely to all integers by

$$h_S(m+1) = h_S(m) - d_S(m). \quad (16.3)$$

The function is strictly decreasing, tends to $+\infty$ toward the left, and tends to $-\infty$ toward the right. Set

$$U_S = \{(m, n) \in \mathbb{Z}^2 : n \geq h_S(m)\}, \quad F(m, n) = (m+1, n), \quad G(m, n) = (m, n+1). \quad (16.4)$$

Lemma 16.2 (Staircase rigidity). *For every $S \subseteq \mathbb{N}$, $F, G : U_S \rightarrow U_S$ are commuting proper injections and their joint graph is connected. The unary components of each map are all one-sided rays. As structures with the two fixed function labels,*

$$(U_S; F, G) \cong (U_T; F, G) \iff S = T.$$

Proof. Since h_S is decreasing, moving one step to the right preserves the defining inequality in (16.4); moving one step upwards does so as well. Both maps are restrictions of coordinate translations, so they are injective and commute. The point $(m, h_S(m))$ has no G -predecessor. At each column the strict drop leaves points with no F -predecessor. Thus both maps are nonsurjective.

Given two points, choose an integer pair coordinatewise above both. It belongs to U_S and is reached from each point by positive horizontal and vertical steps. This proves connectedness of the underlying undirected graph.

Each fixed column is a vertical ray. For a fixed height n , strict monotonicity and the two limits of h_S imply that

$$\{m \in \mathbb{Z} : n \geq h_S(m)\} = \{m \in \mathbb{Z} : m \geq m_n\}$$

for one integer m_n . Hence every horizontal component is a ray too.

Suppose $\phi : U_S \rightarrow U_T$ is a labelled isomorphism. Preservation of each edge gives

$$\phi(z + e_i) - (z + e_i) = \phi(z) - z \quad (i = 1, 2)$$

whenever the edge occurs. Connectedness makes the displacement constant. Thus $\phi(z) = z + (a, b)$ for some integers a, b , and surjectivity implies

$$h_T(m+a) = h_S(m) + b \quad (m \in \mathbb{Z}).$$

Taking consecutive differences gives $d_T(m+a) = d_S(m)$. The drop 3 occurs exactly at $m = 0$ for every code, forcing $a = 0$. The normalization $h_S(0) = h_T(0) = 0$ forces $b = 0$. Finally the drops for $m > 0$ recover every bit of S , so $S = T$. Equality of the codes gives the identity isomorphism in the other direction. $\square$

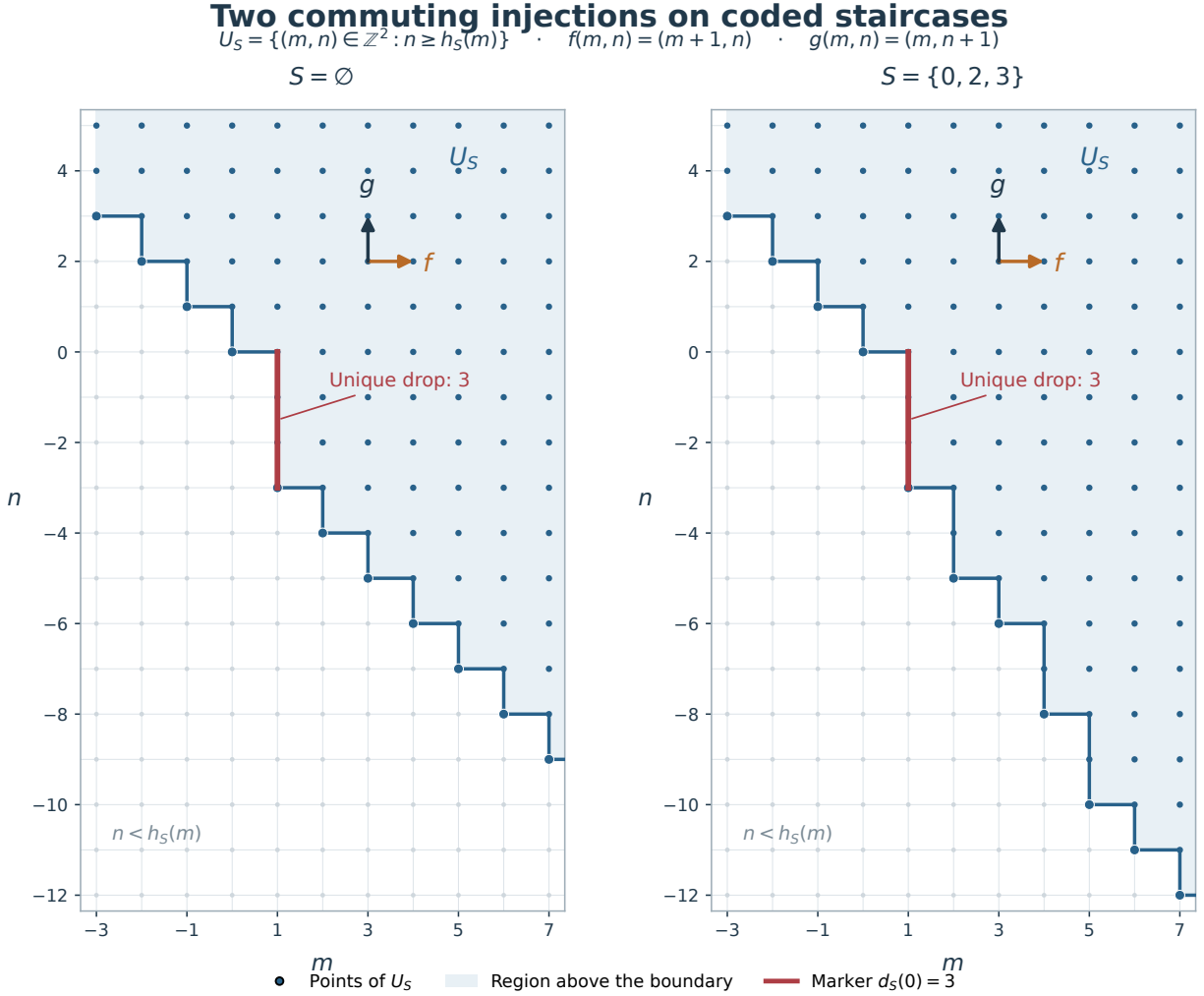


Figure 1: Finite windows in two of the infinite lattice structures U_S . Every lattice point on or above the indicated staircase belongs to U_S . The two coordinate injections commute. The unique height drop of 3 fixes an isomorphism's horizontal translation, while the successive drops of 1 or 2 encode the bits of S . Only the displayed windows are finite; the definitions and the proof concern all of $\mathbb{Z}^2$.

This is a classification of labelled graphs. Swapping the two operation names is not permitted. It is also a statement about countably infinite structures: any fixed finite window sees only finitely many bits, which is why the verification program is explicitly described as a finite check rather than a proof of the continuum claim.

Theorem 16.3 (Complete mixed-signature threshold). *Let $s, t \geq 0$ with $s + t > 0$. For every infinite κ and every A with $|A| \geq \kappa$, universal preservation of expanded Replacement for pairwise commuting s permutations and t additional injections is equivalent to*

$$\begin{cases} \text{cf}(\kappa) > \omega, & t \leq 1, \\ \text{cf}(\kappa) > \mathfrak{c}, & t \geq 2. \end{cases} \quad (16.5)$$

Proof. When $t = 0$, apply [theorem 15.3](#). When $t = 1$, [lemma 16.1](#) gives countably many types. The positive bound follows from [theorem 14.2](#). At uncountable countable-cofinality cutoffs, the distinct-cycle construction remains available because an injection name is allowed to act bijectively. At $\kappa = \omega$, a shift or a ray gives an infinite component. This proves sharpness for $t = 1$.

When $t \geq 2$, there are at most $\mathfrak{c}$ countable structures in the fixed finite language. This proves the positive bound. By [lemma 16.2](#) there are already $\mathfrak{c}$ distinct connected types for two commuting injections. The block construction of [lemma 14.4](#), with every other name the identity, gives failure at each uncountable cutoff with $\text{cf}(\kappa) \leq \mathfrak{c}$. At the finite cutoff one ray suffices, or one may use a single staircase component. This proves the remaining boundary case. $\square$

16.3 Individually preserving maps with a joint failure

Theorem 16.4 (Two commuting proper lifts with incompatible joint Replacement). *Suppose $\kappa > \omega$, $|A| \geq \kappa$, and $\text{cf}(\kappa) \leq \mathfrak{c}$. There are commuting proper injections $f, g : A \rightarrow A$ such that:*

- (i) *each of j_f, j_g is an amenable elementary self-embedding of the base model;*
- (ii) *each one-symbol expansion satisfies full expanded Replacement;*
- (iii) *the expansion naming both maps fails expanded Replacement;*
- (iv) *the fixed objects of either lift are exactly the pure objects.*

Proof. Reserve one staircase code first. Put $\delta = \text{cf}(\kappa)$ and choose distinct codes S_i for $i < \delta$ from the remaining $\mathfrak{c}$ codes, together with infinite $\lambda_i < \kappa$ whose cardinal sum is κ . Partition κ atoms into λ_i copies of U_{S_i} for each i . If $|A| = \kappa$, use all of A for these blocks. If $|A| > \kappa$, partition the remaining atoms into copies of the reserved staircase. This is possible because every staircase is countably infinite and the remainder can have cardinality $|A|$.

Let f, g be the coordinate injections on each copy. They commute and are both proper. Every unary component of either map is a ray. Thus, for each unary reduct, its sole nonempty type block is A , of size at least κ , and its small-type part is empty. The uncountable-cutoff profile criterion proves (ii). For the two-symbol structure, the block associated with S_i has size $\lambda_i < \kappa$, and their union has size κ . The same criterion proves (iii). Part (i) is [proposition 13.4](#).

For (iv), each unary atom map is a disjoint union of rays. There is no nonempty subset $K \subseteq A$ with $f[K] = K$: an atom in such a set would have predecessors of all finite lengths in its ray, contradicting its finite distance from the root. If $j_f(x) = x$, [\(13.2\)](#) gives $f[\ker(x)] = \ker(x)$, so $\ker(x) = \emptyset$. Conversely every pure object is fixed. The argument for g is identical. $\square$

The hypotheses concern a fixed action as well as a universal bound. At a cutoff such as $\aleph_\omega$, some single maps already fail Replacement, but the particular two maps in [theorem 16.4](#) do not. Their joint geometry creates the failure. At $\aleph_1$, the example works in every ambient universe with Choice, since $\aleph_1 \leq \mathfrak{c}$.

17 The exact atom kernel of definable closure

The profile proof gives a uniform bound for unique outputs. For commuting permutations this bound is exact in a strong sense, and the result remains valid when $\kappa = \omega$.

17.1 What is being measured

Fix commuting permutations $\mathbf{p} = (p_1, \dots, p_r)$, let $G = \mathbb{Z}^r$, and let $\bar{a}$ be a finite tuple of internal parameters. Put

$$P = \bigcup_i \ker(a_i), \quad G \cdot P = \{g \cdot b : g \in G, b \in P\}, \quad H(\bar{a}) = G \cdot P \cup D_\kappa(\mathbf{p}).$$

We use the notation

$$\mathcal{K}(\text{dcl}(\bar{a})) = \bigcup \{\ker(y) : y \text{ is uniquely definable from } \bar{a}\}$$

only in the scheme sense of [remark 13.1](#). An equality with $H(\bar{a})$ abbreviates two assertions: every fixed uniquely defining formula has its output kernel contained in $H(\bar{a})$, and every atom of $H(\bar{a})$ occurs in an explicitly exhibited definable object's kernel. No uniform satisfaction predicate is assumed.

An atom occurring in a definable set need not itself be definable. For example, a unique two-element atom block can be definable even though its two atoms are interchanged by an automorphism. The object under study is the kernel of definable objects, not just the atom-valued part of ordinary definable closure.

Theorem 17.1 (Exact definable-kernel formula). *For every infinite κ , every finite tuple of commuting atom permutations, and every finite tuple $\bar{a}$ of internal parameters,*

$$\mathcal{K}(\text{dcl}(\bar{a})) = G \cdot \bigcup_i \ker(a_i) \cup D_\kappa(\mathbf{p}). \quad (17.1)$$

Allowing additional pure parameters does not enlarge this atom kernel.

Proof. Every atom on the right occurs in a definable object. For a fixed group word g , the object $j_g(a_i)$ is definable from a_i . Negative letters are expressed by the unique inverse of the corresponding permutation lift. Its kernel is $g[\ker(a_i)]$. This exhibits every atom of $G \cdot P$ without requiring the entire saturation to be an internal set.

For $K \leq G$, the block

$$B_K = \{b \in A : \text{Stab}_G(b) = K\}$$

is parameter-free definable as a class by [lemma 15.2](#). Whenever $|B_K| < \kappa$, it is an internal set uniquely specified by

$$\text{Set}(y) \wedge \forall b (b \in y \longleftrightarrow \text{At}(b) \wedge \text{Stab}_G(b) = K).$$

The explicit set condition matters when the block is empty: it excludes atoms, which also have no members. Thus every atom of every small block occurs in a parameter-free definable object's kernel, proving the lower inclusion.

No uniquely defined object introduces an atom outside the right side. Suppose y is uniquely defined from $\bar{a}$ and $b \in \ker(y) \setminus H(\bar{a})$. Its orbit C is disjoint from P , and its entire type block is not small.

If $\kappa > \omega$, the saturation $G \cdot P$ has size below κ . The large block of b contains at least κ countable components, so one can interchange C with an isomorphic component avoiding $G \cdot P \cup \ker(y)$. This atom permutation commutes with all the names, fixes $\bar{a}$, and sends b outside the old kernel of y . Its lift contradicts uniqueness. In this argument one does *not* protect all of D_κ pointwise: it may be large, and it is unnecessary to protect it.

Suppose instead that $\kappa = \omega$, so P and $\ker(y)$ are finite. If C is infinite, its transitive abelian action is G/K . Translation by an element of G/K commutes with the entire named action and can send b to any prescribed point of C . Choose a point outside the finite set $\ker(y)$, translate on C , and act identically on every other component. This fixes $\bar{a}$ and changes y , again a contradiction.

If C is finite, its type block must be infinite because $b \notin D_\omega$. Infinitely many copies of its fixed finite type are available. Swap C with a copy avoiding the finitely many components meeting P or $\ker(y)$. This supplies the same contradiction.

Every atom permutation used above fixes all pure objects. Therefore arbitrary additional pure parameters leave the upper bound unchanged, and the lower inclusion already needed none. This proves the final assertion. $\square$

[write] Question 10.1 for commuting permutations, 4 October 2026, batch 93. For commuting permutations the component closure $\text{cl}_p(P)$ of (5.4) is the orbit saturation $G \cdot P$, so [theorem 17.1](#) says that the bound of Part I's Question 10.1 ([Question 10.1](#)), component closure plus D_κ , is exactly the atom kernel of definable closure, at every cutoff. A dated note after [Question 10.1](#) records this and what stays open.

Corollary 17.2 (A single definable bound when the hull is small). *The class $H(\bar{a})$ has a first-order definition from $\bar{a}$. If $|H(\bar{a})| < \kappa$, it is itself one internal definable set of atoms, containing the kernels of all objects definable from $\bar{a}$.*

For $\kappa > \omega$, expanded Replacement holds exactly when the parameter-free definable kernel is smaller than κ .

Proof. The first term of $H(\bar{a})$ is defined by finite-word reachability from the atom kernels of the parameters. To define D_κ uniformly, let a finite integer matrix code a subgroup of G , and ask whether there is an internal set containing exactly the atoms with that stabilizer. An atom belongs to D_κ exactly when it belongs to one of these complete internal blocks. This quantifies over matrices as pure sets and over the individual block witnesses; it does not assume that all witnesses have already been collected. If the resulting class has size below κ , its external extension is an internal set and is uniquely defined by this formula.

With no parameters, [theorem 17.1](#) gives precisely D_κ . The uncountable-cutoff equivalence is now [theorem 14.2](#). For a nonempty parameter tuple, the orbit saturation of its kernel is always small at an uncountable cutoff, so Replacement also makes its full hull small. $\square$

At the finite cutoff the last equivalence requires an additional clause. A single regular infinite $\mathbb{Z}'$ -orbit has empty D_ω and no parameter-free definable atom kernel, yet Replacement fails once an atom is admitted as a parameter and all its iterates are collected. The orbit-finiteness condition in [theorem 14.2](#) is therefore indispensable.

Example 17.3 (Definable kernels can be large without a collecting set). In the countable-cofinality cycle example, each B_n is parameter-free definable. Their union has size κ and cannot be an internal set. Formula (17.1) records exactly this external kernel, while making no assertion that the model contains a set of all parameter-free definable objects. The failed Replacement instance is what would otherwise perform the forbidden collection.

Example 17.4 (Why injections require a correction at ω). Let A be a single countably infinite ray and let one injection be its successor map. At $\kappa = \omega$, its entire ray block is infinite, so $D_\omega = \emptyset$. Nevertheless the root is the unique atom without a predecessor and is parameter-free definable. Every finite successor of the root is then definable. Thus the parameter-free definable kernel is all of A . This disproves the unchanged version of (17.1) for injections and explains the permutation hypothesis in the theorem.

18 A group-theoretic threshold for named automorphisms

Commutation is one possible relation among named automorphisms. A fixed finitely generated group provides a common framework for arbitrary such relations. Let $G = \langle s_1, \dots, s_r \rangle$ and let $\rho : G \rightarrow \text{Sym}(A)$ be an external action. Name the canonical lifts of $\rho(s_1), \dots, \rho(s_r)$ in $M_\kappa(A)$. Relations in G hold for these lifts because canonical lifting respects composition. Neither faithfulness of the action nor finite presentation of G is assumed.

Write

$$\tau(G) = |\{H : H \leq G\} / \text{conjugacy}|. \quad (18.1)$$

This counts conjugacy classes of subgroups, rather than conjugacy classes of elements. It measures how many different transitive atom components the named action can distinguish.

Lemma 18.1 (Transitive component types). *The possible connected-component types for the named generators of G are exactly the transitive G -sets G/H , with H taken up to conjugacy.*

Proof. The component containing a is its orbit Ga . If $H = \{g \in G : ga = a\}$, the map $gH \mapsto ga$ is an equivariant bijection $G/H \rightarrow Ga$. Changing the basepoint conjugates its stabilizer. Conversely, an equivariant bijection $G/H \rightarrow G/K$ taking H to gK identifies stabilizers and therefore gives $H = gKg^{-1}$. A bijection preserving the named generators preserves their inverses and every word in them, so it is exactly a G -equivariant bijection. $\square$

Because G is finitely generated, it is countable. All its transitive components are consequently finite or countably infinite. There is no definability assumption hidden in the use of subgroup types. The pure set of codes for connected structures in the finite generator language, with domain a positive finite ordinal or $\mathbb{N}$, exists in $M_\kappa(A)$, as does its quotient by isomorphism. One can restrict that pure set to the codes satisfying the relations of G . The group, its multiplication table, and a choice of words representing its elements are pure set parameters if needed.

For $\kappa > \omega$, a component code is recognized at an atom by an internal enumeration graph preserving every generator, whose image contains that atom and is closed under incoming and outgoing labelled edges. Such a graph has at most countably many atoms in its kernel. Finite components have finite enumeration graphs even when $\kappa = \omega$. Thus the type-block formula underlying Theorem 14.2 applies without using expanded Replacement to construct its witnesses. This point matters: simply counting abstract subgroup types would not, by itself, supply an internal failed instance of an axiom scheme.

Theorem 18.2 (Exact group robustness threshold). *Let G be a finitely generated group, with a fixed finite generating tuple. Every action of G on A preserves Replacement after the canonical lifts of those generators are named if and only if*

$$\begin{cases} G \text{ is finite,} & \kappa = \omega, \\ \text{cf}(\kappa) > \tau(G), & \kappa > \omega. \end{cases} \quad (18.2)$$

In particular, the answer is independent of the finite generating tuple.

Proof. Suppose first $\kappa > \omega$ and $\text{cf}(\kappa) > \tau(G)$. An action has at most $\tau(G)$ type blocks. The union of its blocks of cardinality below κ therefore has cardinality below κ , so Theorem 14.2 proves preservation.

For the converse put $\delta = \text{cf}(\kappa)$ and assume $\delta \leq \tau(G)$. Choose pairwise nonconjugate subgroups H_i for $i < \delta$, omitting G itself. Omitting one type does not reduce the available infinite cardinality. Choose infinite cardinals $\lambda_i < \kappa$ with

$$\sum_{i < \delta} \lambda_i = \kappa.$$

For regular κ , take $\delta = \kappa$ and every $\lambda_i = \aleph_0$; for singular κ , take a cofinal family of infinite cardinals below κ .

On a subset of A of cardinality κ , place λ_i disjoint copies of G/H_i for each $i < \delta$. Fix the remaining atoms. The selected type block B_i has size λ_i , since each G/H_i is nonempty and countable, and the selected blocks together have size κ . The remainder has trivial type and cannot enlarge any B_i . Thus the small-type union violates the criterion in Theorem 14.2.

Explicitly, a pure sequence of the selected type codes is an internal parameter. The formula sending i to the set of all atoms having the i th type has a unique internal value B_i for every $i < \delta$. Replacement for this function would produce a range set whose kernel contains their union of size κ . This is the required failed instance, on the pure internal domain δ .

At $\kappa = \omega$, a finite group has only finitely many transitive types, all finite. Its union of finite type blocks is finite, so the finite-cutoff clause of Theorem 14.2 applies. If G is infinite, place one regular G -orbit on a countable atom subset and fix the remainder. From an atom in that orbit, endpoints of finite words in the generators and inverses are uniquely definable. Every finite path graph exists internally. Replacement on the pure set of words would collect the entire infinite orbit into an internal atom set, impossible at the finite-kernel cutoff. $\square$

[write] **The finite cutoff in the repository, 4 October 2026, batch 93.** The clause for $\kappa = \omega$ also follows from Part VIII of birthday-cutoffs-and-hereditary-sets, written after this manuscript's pin: by its Proposition 62.1 (hset:ns:prop:finitewords) the generator names define the uniform action, and its Theorem 58.1 (hset:ns:thm:full) then requires finite orbits and a finite exceptional set, which every action has exactly when G is finite. The clause for $\kappa > \omega$, the threshold $\text{cf}(\kappa) > \tau(G)$, is new here.

18.1 The three possible type counts

Proposition 18.3 (Cardinality trichotomy). *For a countable group G , the cardinal $\tau(G)$ is finite, $\aleph_0$, or $\mathfrak{c}$.*

Proof. Represent a subgroup by its characteristic function in 2^G . The subgroup conditions are closed: failure to contain the identity or to be closed under products and inverses has a finite witness. Thus the space of subgroups is a closed subspace of the compact metrizable space 2^G . If uncountable, it

has a nonempty perfect subset and therefore cardinality $\mathfrak{c}$. For clarity, the perfect-set step follows by repeatedly splitting basic clopen neighborhoods around distinct condensation points; the resulting binary tree embeds $2^{\mathbb{N}}$. Only countably many points can fail to be condensation points, since the space has a countable base. Every conjugacy class of subgroups is countable, being an orbit of G . Consequently an uncountable subgroup space has $\mathfrak{c}$ conjugacy classes as well. The remaining possibilities are finite or countably infinite. $\square$

For an infinite finitely generated abelian group the count is $\aleph_0$: every subgroup is finitely generated, so a countable collection of finite tuples lists them, and infinitely many finite-index subgroups give distinct types. The same upper-bound argument applies whenever every subgroup of a countable group is finitely generated. At the other extreme, the free group $F(a, b)$ has $\mathfrak{c}$ types. The elements $a^n b a^{-n}$, $n \in \mathbb{Z}$, freely generate a subgroup with a countable free basis. Indeed, after combining adjacent equal basis indices, expansion of a nonempty reduced word leaves nonzero powers of a between consecutive nonzero powers of b , so it cannot reduce to the identity. Distinct subsets of this basis generate distinct subgroups, giving $\mathfrak{c}$ subgroups; quotienting by the countable conjugacy classes preserves that cardinality.

Remark 18.4 (Infinite groups with finitely many types). The finite alternative cannot be identified with finiteness of G . Smith records finitely generated infinite simple groups constructed by Ivanov and Ol'shanskii in which all proper nontrivial subgroups are conjugate [11, p. 69]. Such a group has $\tau(G) = 3$. We use this published existence statement and do not reproduce that construction. Theorem 18.2 then says that every named generator action preserves Replacement at every uncountable cutoff, while a regular orbit still defeats it at $\kappa = \omega$. This is why the finite-cutoff clause must be stated separately.

[write] An imported existence theorem, 4 October 2026, batch 93. This remark is not proved here: it imports the existence of a finitely generated infinite group all of whose proper nontrivial subgroups are conjugate, a construction of Ivanov and Ol'shanskii [12] that the manuscript read only through Smith [11]. At the write, the first page of [11], freely visible on the publisher's site, was checked: it attributes to Ivanov and Ol'shanskii finitely generated infinite simple groups with all proper nontrivial subgroups conjugate. The primary source (paywalled) was not consulted, by the manuscript or at the write. Given the existence statement, $\tau(G) = 3$ is immediate: the classes are $\{1\}$, G , and the single class of proper nontrivial subgroups, which is nonempty because an infinite simple group is not cyclic. What depends on the import is only the claim that the two clauses of theorem 18.2 genuinely differ, that is, that an infinite finitely generated group can have finite $\tau(G)$; theorem 18.2 and proposition 18.3 are proved without it. Question 23.13 records the missing primary check.

The topology of subgroup spaces has a much deeper established theory, including a complete analysis for countable abelian groups in [10]. Only the elementary cardinality fact above is needed here.

These results translate group structure into preservation properties of a set-theoretic scheme. They do not classify groups with countably many subgroups or claim to solve the existing classification problems about subgroup spaces. Their role is to identify exactly which invariant of a fixed finitely generated action controls this family of kernel models.

19 Uniformly evaluating an infinite family of automorphisms

There are two different ways to name an action of a countable group. In the first, the language has a separate unary symbol J_g for each $g \in G$. Every first-order formula still contains only finitely

many of these symbols. In the second, one binary symbol E uniformly evaluates the action:

$$E(g, x) = j_{\rho(g)}(x) \quad (g \in G). \quad (19.1)$$

Set $E(g, x) = x$ when $g \notin G$ to make it total, and allow the pure set G as an internal parameter. Now a single formula can quantify over all $g \in G$. Although the individual maps in the two descriptions are identical, their Replacement schemes need not agree.

Proposition 19.1 (Separate names for a locally finite group). *If G is locally finite, every action of G preserves Replacement in the language of separate unary names at every infinite cutoff. For countable G , the converse holds for universal preservation at $\kappa = \omega$.*

Proof. A formula uses finitely many names. They generate a finite subgroup when G is locally finite, so Theorem 18.2 applies to that formula. Conversely, if a finite subset generates an infinite subgroup K , a regular G -orbit on countably many atoms restricts to an infinite K -orbit. The finite-word endpoint instance using those finitely many names fails at $\kappa = \omega$. $\square$

Proposition 19.2 (Uniform group profile). *For a countable group acting with a uniform evaluation name, the component-profile criterion of Theorem 14.2 holds with components equal to full G -orbits and types equal to conjugacy classes of stabilizers. Consequently the universal robustness criterion (18.2) holds for every countable group in this language, without a finite-generation assumption.*

Proof. Use pure codes for actions of the pure set G on a positive finite ordinal or $\mathbb{N}$. An internal enumeration graph h recognizes a component by requiring $h(g \cdot d) = E(g, h(d))$ for every $g \in G$ and every code element d . Transitivity of the code and inclusion of the chosen atom identify the image with its whole orbit. This is one formula precisely because evaluation is uniform. The graph is internal at uncountable cutoffs, and for finite orbits at the finite cutoff.

The necessity argument therefore collects the small type blocks just as before. At $\kappa = \omega$, the uniformly defined function $g \mapsto E(g, a)$ first forces every orbit to be finite. For sufficiency, saturate the kernels of the input set and parameters under G , and adjoin the small-type part. This protected union has size below κ : countable saturation preserves that bound when $\kappa > \omega$, while saturation of finitely many atoms is finite when all orbits are finite. A fresh equivariant component swap fixes the protected union and changes any purported unique output using an atom outside it. Its lift preserves E , so uniqueness confines all output kernels to the protected union. External Replacement then produces an internal range. Finally the sharp type-block construction in Theorem 18.2 uses only countability of G and applies unchanged. $\square$

[write] Prior in the repository at the finite cutoff, 4 October 2026, batch 93. At $\kappa = \omega$ both propositions are in Part VIII of birthday-cutoffs-and-hereditary-sets, written after this manuscript's pin: [proposition 19.1](#) is its finite-symbol principle, Proposition 62.2 (`hset:ns:prop:language`), together with Corollary 58.2 (`hset:ns:cor:finitigroup`); [proposition 19.2](#) is its Theorem 58.1 (`hset:ns:thm:full`), stated there for a uniformly named action of an arbitrary pure set-sized group, its exceptional set X_{Act} being D_ω once all orbits are finite. The proofs are independent. For $\kappa > \omega$ both propositions are new here; [proposition 19.2](#) answers in part that Part's Question 65.2 (`hset:ns:q:kernelbounds`), whose dated note lists "actions of arbitrary pure groups at uncountable κ " as open: for countable groups and full Replacement the criterion is the profile criterion with full orbits, and the universal threshold is $\text{cf}(\kappa) > \tau(G)$. Uncountable groups and the bounded levels $\lambda < \kappa$ of that question remain open.

19.1 A separation using only two-atom orbits

Take

$$G = \bigoplus_{n < \omega} C_2,$$

coded as the pure set of finite subsets of $\mathbb{N}$ with symmetric difference. This group is locally finite. For each nonempty $S \subseteq \mathbb{N}$, its nonzero character

$$\chi_S(F) = |F \cap S| \pmod{2} \quad (19.2)$$

defines a transitive action on $\{0, 1\}$ by $F \cdot \varepsilon = \varepsilon + \chi_S(F) \pmod{2}$. Distinct characters have distinct kernels: an index-two subgroup has a unique nonzero quotient character to C_2 . Because G is abelian, these kernels are never identified by conjugacy. There are therefore $\mathfrak{c}$ different two-point transitive types.

Theorem 19.3 (An amenable uniform name can change Replacement). *For every infinite κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, there is an action of $G = \bigoplus_{n < \omega} C_2$ on A , all of whose orbits have size at most two, such that:*

1. *naming every individual lift separately preserves Replacement;*
2. *naming the uniform evaluation function E violates Replacement;*
3. *E is amenable: its restriction graph on every internal set of input pairs belongs to $M_\kappa(A)$.*

For $\text{cf}(\kappa) > \mathfrak{c}$, every uniformly named action of this group preserves Replacement.

Proof. At $\kappa > \omega$, let $\delta = \text{cf}(\kappa)$ and choose distinct nonempty codes $S_i, i < \delta$. Choose small infinite multiplicities λ_i with sum κ . Put λ_i copies of the two-point action of χ_{S_i} in a block B_i , and fix the remaining atoms. Then $|B_i| = \lambda_i < \kappa$, whereas the union of the B_i has size κ . At $\kappa = \omega$, take one copy of each of countably many distinct nonzero characters; each B_i then has size two.

In the uniform language the formula

$$\text{At}(a) \wedge \forall g \in G (E(g, a) = a \iff \chi_S(g) = 0) \quad (19.3)$$

recognizes the type of χ_S . Its arithmetic character computation is pure and uniformly definable. A pure sequence of the selected S_i therefore defines the unique internal block B_i from i . Any Replacement range for these values would have a kernel reaching the cutoff. This proves failure even though no orbit is infinite. Preservation for the separate-name expansion follows from Proposition 19.1.

For amenability, let w be an internal set of input pairs and put $P = \ker(w)$. The kernel of any value $E(g, x)$ appearing on w is contained in the saturation GP . Since every orbit has at most two atoms, $|GP| \leq 2|P| < \kappa$, with the evident finite interpretation. The externally constructed restriction graph has kernel contained in GP and is therefore internal. Finally $\tau(G) = \mathfrak{c}$ by the two-point types and the general continuum upper bound, so Proposition 19.2 gives the asserted positive threshold. $\square$

The witness here does more than expose an unavailable infinite orbit. All individual named maps are commuting involutions, every full orbit has at most two atoms, and the uniform evaluation class is itself amenable. The failed instance uses the ability of one formula to test all group indices and distinguish their stabilizers. Replacing infinitely many separate names by a uniform evaluator is thus a substantive change to an axiom scheme.

[write] **Relation to Part VIII of the surreal report, 4 October 2026, batch 93.** Theorem 62.3 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:thm:evaluator`), written after this manuscript's pin, makes the same separation for the same group at $\kappa = \omega$: countably many separately named involutions keep Replacement, and one uniform evaluator $T(i, a, b)$ destroys it. Its witness is the regular action of $\bigoplus_{\omega} C_2$ on atoms labelled by the group, whose orbit is infinite; the evaluator with one atom parameter then collects infinitely many single atoms, so already R_1 (Replacement for outputs with at most one atom) fails.

[Theorem 19.3](#) differs in four ways: every orbit has at most two atoms, so the failure comes from distinguishing stabilizers rather than from an infinite orbit; it holds at every κ with $\text{cf}(\kappa) \leq \mathfrak{c}$, not only at ω ; the evaluator is amenable; and the converse for $\text{cf}(\kappa) > \mathfrak{c}$ is included. At $\kappa = \omega$ the two witnesses are also separated by the graded levels of that Part: for the witness above, every orbit is finite, and at most one atom (a fixed atom, when it is the only fixed one) has an orbit of size at most one under the centralizer of the action, because the transposition of a two-atom orbit commutes with the action; so by its Theorem 59.1 (`hset:ns:thm:graded`; its language is interdefinable with the evaluator E by its Proposition 62.4) R_1 holds, while R_2 fails, since each of the infinitely many two-atom orbits is the only orbit of its type and so has centralizer orbits of size two. This last observation is the write's, by that theorem, and is not in either manuscript.

Proposition 19.4 (Uniform evaluation for a finitely generated group). *If G is finitely generated, its uniform evaluation function is definable from the finitely many generator names and a pure parameter. Adjoining that evaluator therefore does not change whether Replacement holds.*

Proof. Choose a pure function assigning to every $g \in G$ a finite word in the generators and their inverses representing g . To define $E(g, x)$, ask for the endpoint of a finite path starting at x and following that word. A positive letter applies its named lift; a negative letter requires the named lift of the next value to equal the previous value. These steps are unique because the lifts are automorphisms.

Every such finite path graph exists internally. Its finitely many values have kernels which are images of $\ker(x)$ under group elements, so their union has size below κ , even when κ is singular or equals ω . The defining formula thus uses only the finite generator language and the pure word-selector parameter; it assumes no expanded Replacement. Substituting this definition into an evaluation-language Replacement instance translates it to an instance in the generator language. Conversely each generator is an evaluation at one fixed pure group element. $\square$

This proposition specifies the boundary of the preceding counterexample. The issue is not merely that an index variable occurs in the notation. For finitely generated actions a finite path can evaluate every index using fixed named operations, and its existence is already guaranteed by the kernel bound. For $\bigoplus_{n < \omega} C_2$, no finite family of group elements generates the full action; a formula in separate names has access only to a finite subgroup, while the uniform predicate (19.3) compares all its indices at once.

[write] **Added 4 October 2026, batch 93.** At $\kappa = \omega$, [proposition 19.4](#) is Proposition 62.1 of Part VIII of birthday-cutoffs-and-hereditary-sets (`hset:ns:prop:finitewords`), stated there for the words of a pure free group on finitely many named permutations and written after this manuscript's pin; here it holds at every cutoff.

19.2 Finite orbits and the profinite completion

There is an exact version of this phenomenon for any countable group. Define

$$\tau_{\text{fin}}(G) = |\{H \leq G : [G : H] < \omega\} / \text{conjugacy}|.$$

Corollary 19.5 (Universal preservation for finite-orbit actions). *Restrict to actions of a countable group G whose every orbit is finite, and name their uniform evaluation function. Every such action preserves Replacement exactly under the conditions*

$$\begin{cases} \tau_{\text{fin}}(G) \text{ is finite,} & \kappa = \omega, \\ \text{cf}(\kappa) > \tau_{\text{fin}}(G), & \kappa > \omega. \end{cases}$$

At the finite cutoff the condition is equivalent to finiteness of the profinite completion $\widehat{G}$.

Proof. The uniform profile proof now uses only finite transitive types. At uncountable cutoffs its upper bound and sharp block construction apply with $\tau_{\text{fin}}(G)$. At the finite cutoff, finitely many types give a finite union of finite blocks. Infinitely many types allow one copy of each of countably many distinct nontrivial types, whose finite blocks have infinite union.

A finite-index subgroup has finitely many conjugates, since its normalizer contains it. Hence finite $\tau_{\text{fin}}(G)$ is equivalent to having finitely many finite-index subgroups. Intersect their normal cores to obtain a finite-index normal subgroup R contained in every such subgroup. Every finite quotient factors through G/R , so $\widehat{G} = G/R$ is finite. Conversely, each finite-index subgroup is the inverse image of an open subgroup of $\widehat{G}$: its finite coset action extends to the profinite completion, and one takes the stabilizer of its distinguished coset. If $\widehat{G}$ is finite there are only finitely many such subgroups. $\square$

For a finitely generated group there are only finitely many actions on each fixed finite labelled set: the images of finitely many generators determine the action. Thus $\tau_{\text{fin}}(G)$ is at most countable. The continuum-sized family of two-point types above relies on infinitely many generators and their uniform evaluation. This explains precisely why its language distinction has no counterpart for a fixed finite generating tuple.

20 Collection and reflection in the commuting cases

The preceding results concern Replacement. The relation to Glazer and Yao's research becomes clearer when Collection and reflection are kept distinct. In the present external-choice models, the general Collection and full-reflection criteria were already established in [7, Theorems 7.5 and 7.7]. We record and prove the consequence for the countable-type commuting signatures.

Theorem 20.1 (Collection and reflection for at most one injection). *Fix a finite nonempty commuting tuple of injections in which all but at most one map are permutations. In its canonical expansion of $M_\kappa(A)$, the following are equivalent:*

- (i) *full expanded Collection;*
- (ii) *for every finite family Φ of expanded formulas and every internal set w , there is an internal transitive set t with $w \subseteq t$, closed under the named functions, such that every formula in Φ is absolute between t and the full model for all parameter tuples from t ;*
- (iii) *κ is an uncountable successor cardinal.*

This is a consequence of the prior Collection/reflection criterion and the type classification proved here.

Proof. Failure at a limit cutoff. Suppose κ is a limit cardinal, with ω included. Choose a pure set w of ordinals below κ whose cardinalities are unbounded below κ . For ω , take $w = \omega$. For each $\alpha \in w$ there is an internal graph of an injection from α into the atoms: it uses only $|\alpha| < \kappa$ atoms. A set collecting such graphs for all $\alpha \in w$ would have a kernel of size at least $|\alpha|$ for every $\alpha \in w$, contrary to the cutoff. This is a failed base-language Collection instance, so adding names cannot repair it.

A common reservoir at a successor cutoff. Write $\kappa = \nu^+$, with ν infinite. There are at most countably many component types, and every component is countable. Close the kernels of an internal input domain w and its parameters under components; this uses at most ν atoms. Add every entire type block of size at most ν . There are only countably many such blocks, so the resulting component-closed set P^* still has size at most ν .

For each remaining realized type, choose ν components disjoint from P^* . Such a type has at least ν^+ components. Their union with P^* is a fixed component-closed reservoir H of size at most ν . If y is any internal witness, its kernel meets at most ν components outside P^* . For each type, match those components injectively into its reservoir. Because the full component family has size at least ν^+ , the partial matching extends to a permutation of that family; choose the corresponding component isomorphisms. The resulting atom permutation commutes with all names and fixes P^* pointwise. It carries y to a witness whose kernel lies in H while fixing every input and parameter.

For a fixed total relation on w , external Collection and Choice now supply witnesses all supported on the same H . Their collected set is internal. This proves full expanded Collection independently of the cardinality of the pure part of w .

Reflection. For a finite family of formulas, include its subformulas, expressing universal quantifiers through negation and existence. Construct externally increasing sets

$$X_n = V_{\alpha_n}(H_n) \quad (n < \omega),$$

where H_n is component-closed of size at most ν , the α_n increase, and the initial stage contains the desired parameters and elements of the prescribed internal set. Given X_n , the reservoir construction over H_n compresses all needed witnesses while fixing their parameter tuples. There are only set-many tuples in X_n and finitely many existential subformulas. External Collection gives a common rank bound, producing H_{n+1}, α_{n+1} such that each true existential instance over X_n has a witness in X_{n+1} .

The union $t = \bigcup_n X_n$ is transitive, is closed under each named function, and has kernel of size at most ν , so it is internal. Induction on the finite collection of formulas proves absoluteness between the full model and t . The existential step is precisely the witness property; the atomic step uses transitivity and closure under the named functions.

Conversely full reflection implies Collection by reflecting a totality assertion and its matrix. For a domain w and parameter tuple p , prescribe an internal set containing w itself and all the entries of p , so that these objects belong to the reflecting set t . Transitivity then puts every element of w in t . The witnesses supplied in t are genuine witnesses by absoluteness, and t itself is an internal collecting set. This completes the equivalence. $\square$

Remark 20.2 (The reservoir must grow). Reflection uses increasing atom reservoirs as well as increasing rank bounds. A single $V_\alpha(H)$ can contain its whole atom supply H as one of its elements, whereas the full kernel model has no internal set of all atoms. The growing-reservoir proof avoids incorrectly treating one fixed atom supply as a reflecting universe.

These results give several distinct possibilities. At an uncountable successor cutoff, every commuting-permutation expansion has full Replacement, Collection, and the specified full reflec-

tion. At a limit cardinal of uncountable cofinality, every such expansion still has Replacement, but Collection and full reflection fail. At a limit cardinal of countable cofinality, Replacement depends on the specific action's small-block profile, and Collection always fails.

This does not recover or strengthen the choiceless separation theorem of Glazer and Yao. Their work varies choice and reflection principles in a broader collection of models. Here the ambient construction uses Choice, and the new classification varies the named action and the atom cutoff. The connection is the common role of atom symmetries in deciding whether witnesses can be confined to one internal set.

21 Forcing stability for a fixed finite action

The universal thresholds quantify over all available actions, so they can change when a forcing extension supplies new actions. For an action already present in the ground universe, the profile theorem yields a stronger stability statement. This answers the fixed-action part of the forcing question proposed in the repository [7, Question 10.10] under an explicit hypothesis on the cutoff.

Theorem 21.1 (Preserving the cutoff preserves the verdict). *Let $V \subseteq W = V[\mathcal{G}]$ be a set-forcing extension of the ambient universe. Suppose A , an infinite cardinal κ , and a fixed finite tuple f of injections on A belong to V , with $|A|^V \geq \kappa$. Assume that the ground ordinal κ remains a cardinal in W . Form the kernel model and the canonical lifts separately in V and in W . Then the ground expansion satisfies expanded Replacement if and only if the rebuilt expansion in W does.*

Proof. The atom set and the named maps have the same elements and values in both universes. A component is determined by finite paths through these values, so each component is the same ground set in V and W . It is countable in V .

We next show that the partition into component types does not change. Fix two ground components and ground enumerations of them. Their labelled isomorphisms correspond to infinite branches through a ground tree of finite partial isomorphisms, requiring at the n th stage that the first n enumerated elements on both sides be covered. A ground isomorphism remains one in W . If there is no ground isomorphism, this tree is well-founded in V and has an ordinal-valued rank function there. The same rank function remains valid in W and precludes a new infinite branch. Thus no two previously different component types become isomorphic. Indexing types by the actual partition of components, rather than by all possible pure codes, gives the same blocks B_i in both universes.

For any ground set X , the assertion $|X| < \kappa$ is absolute under the stated cardinal hypothesis. A ground injection of X into an ordinal below κ persists and proves the forward direction. If $|X|^V \geq \kappa$, a ground injection of κ into X persists and, because κ is still a cardinal, prevents $|X|^W < \kappa$.

Consequently the family of small blocks B_i is unchanged, its union D_κ is the same ground set, and the smallness of that union is unchanged. At $\kappa = \omega$, the finiteness of every component and of D_ω is also absolute. Apply [theorem 14.2](#) in the two universes. The models have different pure sets in general, but the criterion applies to all formulas and all internal parameters in each rebuilt model, which is exactly the required equivalence. $\square$

[write] **Question 10.10, fixed-action part, 4 October 2026, batch 93.** “Question 10.10” of [7] is [Question 10.10](#) of Part I; a dated note there records this answer and what remains open ([Question 23.11](#)).

Remark 21.2 (What can still change). The theorem permits a change of $\text{cf}(\kappa)$ as long as κ remains a cardinal. If such a change crosses a universal threshold, counterexample actions available in the extension need not be ground actions. The theorem does not assert that the old proper-class model is unchanged, that each old formula has the same truth value, or that Collection is absolute under arbitrary cardinal collapses below κ . For the countable-type commuting signatures of [theorem 20.1](#), preserving whether κ is a successor cardinal additionally preserves Collection and the stated reflection scheme; forcing that preserves all cardinals is sufficient.

22 Formalization, verification, and proof dependencies

22.1 How the argument fits the existing ProveIt formalization

The inspected repository contains a formula-indexed ZF development at `SetTheory/ZF/Lean/ZF/Zf.lean`. Its declarations include `Sep_form`, `Func_form`, `Image_form`, `Repl_form`, `ZFax`, and `ZFprov` [8]. They provide a natural starting point for distinguishing a base formula language from the enlarged language used by a scheme. They do not already formalize the urelement class models, component profiles, or theorems of this article.

A useful formalization should make the signature an explicit input. A proof of Replacement for formulas in $\{\in, \text{At}\}$ must not automatically become a proof for formulas in $\{\in, \text{At}, J_1, \dots, J_r\}$. Similarly, an interface for amenability should assert the existence of individual restriction graphs and should not quietly include expanded Replacement.

Milestone	Mathematical content to certify
Syntax and interpretation	Separate atom and set predicates; finite function signatures; relativization of each schema instance.
The kernel model	Tagged universe, transitive closure, kernel operations, closure under subsets, and the fresh-atom proof of base Replacement.
Finite paths and lifts	Recursive lift, preservation of kernels, extension of a small partial injection to an atom permutation, and internal finite path graphs.
Component types	Pure countable codes, labelled isomorphisms, uniform recognition, and the fresh-component swap.
Commuting normal forms	Subgroups of $\mathbb{Z}'$, the fiber quotient for one injection, holonomy on cycles, and rigidity of the coded staircases.
Schema certificates	An explicit input domain, functional formula, and cutoff-violating kernel for each negative example.

The core mathematical proof is amenable to modular formalization: finite combinatorics establishes the type families, elementary cardinal arithmetic gives their possible unions, and the kernel model converts those unions into schema instances. The hard formal boundary is semantic bookkeeping for a definable proper-class structure. An eventual formalization may instead work over externally quantified set models with explicit satisfaction and transport each fixed instance. Either route must state its metatheory rather than conceal a truth predicate.

22.2 The supplied computational checks

The file `verify_examples.py` performs deterministic finite checks using integer arithmetic and finite sets only. Its recorded run passes 347,278 explicit checks. These cover all 256 eight-bit staircase prefixes continued by zeros, boundary decoding, coordinate closure and commutation, translated finite boundary shadows, all 15 subgroups of $(\mathbb{Z}/4\mathbb{Z})^2$, quotient-action stabilizers and equivariant maps, and all 31 nonzero characters of $(\mathbb{F}_2)^5$.

The file `make_figure.py` regenerates [figure 1](#). The picture is an exact integer-lattice visualization of the defining inequalities; it is not generated imagery. Both the vector figure and a high-resolution preview are included.

The role of these checks is limited and useful. They test the explicit finite formulas underlying the constructions and make arithmetic or indexing errors reproducible. They do not prove that all infinite staircase codes are distinct, do not verify any infinite-cardinal argument, and do not certify any instance of set-theoretic Replacement. The mathematical proofs of those claims are supplied in the text. No Lean, Coq, or other proof-assistant verification is claimed.

[write] Shipped layout and replay, 4 October 2026, batch 93. In ProveIt the two programs are `code/02-commuting-actions-verify_examples.py` and `code/02-commuting-actions-make_figure.py`, the recorded run is `data/02-commuting-actions-verification_output.txt`, and the figure is `figures/02-commuting-actions-staircases.pdf` with the preview `figures/02-commuting-actions-staircases.png`. Both programs were rerun on scratch copies at the write (Section 12.4); the report’s README gives the commands, which never write into this directory.

22.3 Dependency audit

Conclusion	Essential input	Boundary to preserve
Commuting threshold	Finite descriptions of subgroups of $\mathbb{Z}^r$; type profile	Countability of types is used with cofinality, not mere size of κ .
One-injection threshold	Injective map on permutation orbits; untwisting of fiber maps	Induced fiber maps are bijections even though the global map may not be.
Two-injection lower bound	Rigid staircase boundary; block multiplicities	Complete infinite labelled types are distinct; finite shadows are only checks.
Exact definable kernel	Parameter-free subgroup definitions and transitive translations	Kernels of definable objects differ from individually definable atoms.
General group threshold	Coset-action classification and uniformly recognized codes	Subgroups are counted up to conjugacy; the finite-cutoff case is separate.
Uniformity separation	Infinitely many index-two stabilizers; finite reducts	A binary evaluator permits index quantification unavailable to a fixed finite list of unary names.

Conclusion	Essential input	Boundary to preserve
Collection and reflection	Compression into growing component reservoirs	Arbitrary witnesses require a reservoir argument; uniqueness alone gives only Replacement.

23 Further research questions and topics

The questions below specify where the proved classifications end. They are proposed research tasks, not a claim that every formulation is a longstanding published open problem. The commuting-permutation question itself is no longer included as an open task in the setting of this article.

Question 23.1 (Atom-valued definable closure for nonabelian actions). For a finitely generated nonabelian group G , determine the atoms that are individually definable from specified parameters, as distinct from the kernel of definable objects. The equivariant automorphism group of G/H is controlled by $N_G(H)/H$. How does that normalizer quotient interact with the internal availability of complete orbit blocks? A useful first target is the infinite dihedral group, where orbit and stabilizer data remain explicit.

Question 23.2 (Exact finite-kernel formulas for several injections). Develop the analogue of [theorem 17.1](#) when two or more commuting injections are named and $\kappa = \omega$. The ray in [example 17.4](#) shows that boundary points create definable atoms absent from the small-block term. Which finite layers or distinguished boundary configurations account for all such atoms in the staircase families? Can the result be stated through a finite-boundary invariant of a cancellative monoid action?

Question 23.3 (Relations among noninvertible generators). Classify the connected-type cardinality for injective actions of specified finitely presented commutative monoids. The free case $\mathbb{Z}^s \times \mathbb{N}^t$ is settled here. Relations such as $f^m = g^n$ can collapse two apparent directions. Determine when such relations force countably many types and when a staircase or another real-coded family survives.

Question 23.4 (Using finer subgroup-space invariants). The theorem uses only $\tau(G)$, but subgroup spaces have much finer topology and descriptive complexity; for countable abelian groups see [\[10\]](#). Can Cantor–Bendixson rank or a natural complexity bound on stabilizer codes control the syntactic complexity of failed Replacement instances? The first goal should be a precise theorem for a class of groups with already classified subgroup spaces.

Question 23.5 (Decidability from a finite presentation). For a finitely presented group G , what is the complexity of determining which of the finite, countable, and continuum type-count cases occurs? More realistically, find effective sufficient certificates for a countable bound, a perfect family of subgroup types, or a selected bad multiplicity profile. Any undecidability claim should be proved by an explicit reduction rather than inferred from the general difficulty of group theory.

Question 23.6 (Separate names for groups that are not locally finite). Classify universal Replacement preservation for countable groups in the language of separate names. Local finiteness gives unconditional preservation, and finite generation is settled by [theorem 18.2](#). Between these cases, every formula sees a finitely generated subgroup, but only actions that extend to the whole group are available. Determine whether an invariant of extendible subgroup actions gives a sharp general criterion.

Question 23.7 (Infinitary formulas). Replace ordinary first-order syntax by a precisely specified infinitary logic and re-evaluate the separate-name/uniform-evaluation distinction. For example, can a countable conjunction uniformly distinguish the index-two stabilizers used in [theorem 19.3](#)? The treatment must separately specify formula codes, satisfaction, the allowed parameters, and which Replacement or Collection scheme is being asserted.

Question 23.8 (How much ambient Choice is necessary?). The present proofs use external Choice for cardinal arithmetic, families of isomorphisms, and witness selection after external Collection. Isolate these uses in the commuting case. Rooted finitely generated orbits have explicit word enumerations, but selecting and matching components still needs analysis. Establish the weakest natural choice hypotheses under which the exact threshold and the definable-kernel formula survive.

Question 23.9 (Atom ideals beyond cardinal cutoffs). Replace the condition $|\ker(x)| < \kappa$ by membership in an atom ideal I . Seek conditions on I and the action which are both necessary and sufficient for expanded Replacement. Closure under orbit saturation, closure under unions of recognizable small blocks, and the ability to transport witnesses into fresh copies are distinct requirements and should not be merged into one unexamined symmetry assumption.

Question 23.10 (Complexity of a failed instance). Find the least formula complexity needed to witness the negative examples. The type selector quantifies over whole blocks, while the staircase construction has a directly readable boundary. Can a lower-complexity formula uniformly recognize each coded block? The answer should distinguish a fixed code as parameter, a pure sequence of codes, and a parameter-free formula.

Question 23.11 (Forcing beyond preservation of the cutoff). [Theorem 21.1](#) settles the fixed-action question when the original cutoff remains a cardinal. Analyze what happens when that ordinal is collapsed and the model is rebuilt using its new cardinality as the cutoff. Find exact examples in which Replacement changes, and determine which changes of block sizes cause the transition. Separately, when cofinality changes but the cutoff remains a cardinal, identify how new actions witness any change in a universal threshold.

Question 23.12 (Formal schema independence certificates). Build a reusable ProveIt interface certifying a definable model, its valid scheme instances, and one explicit failed instance in an expanded signature. Begin with the distinct-period example, then certify the two-point uniformity separation. This would provide a concrete test that the formal infrastructure keeps elementary lifts, amenability, base Replacement, and expanded Replacement separate.

[write] **A question added at the write, 4 October 2026, batch 93.** ProveIt’s standing rule for unproved claims (docs/incoming/README.md, Section 4, item 11) moves the one imported, unproved premise of this Part here.

Question 23.13 (A primary check of the Ivanov–Ol’shanskii groups). *Source:* [remark 18.4](#), which credits Ivanov and Ol’shanskii [12] through Smith [11, p. 69]. *Claim:* there is a finitely generated infinite group G all of whose proper nontrivial subgroups are conjugate; then $\tau(G) = 3$, so the finite-cutoff clause “ G finite” of [theorem 18.2](#) and the clause “ $\text{cf}(\kappa) > \tau(G)$ ” read at $\kappa = \omega$ (that is, $\tau(G)$ finite) differ. *Argument available:* none beyond Smith’s attribution (checked on the first page of [11]); the title of [12] names its method, graded diagrams in combinatorial group theory. *Missing:* a check of the construction in the primary source, or in a textbook account of it, confirming finite generation, infinitude and conjugacy of all proper nontrivial subgroups; neither the manuscript nor the write consulted [12]. Short of that, is there a more accessible infinite finitely generated group with finitely many conjugacy classes of subgroups?

[write] **Status of the questions, 4 October 2026, batch 93.** None of these thirteen questions is answered in ProveIt. Several continue questions of Part I and of Part VIII of birthday-cutoffs-and-hereditary-sets (hset:ns:): [Questions 23.1](#) and [23.2](#) continue [Question 10.1](#) beyond commuting permutations; [Question 23.7](#) meets [Question 10.6](#) (uncountably many names and infinitary syntax); [Question 23.8](#) is [Question 10.8](#) (removing ambient choice) for the commuting case; [Question 23.9](#) is [Question 10.7](#) and Part VIII’s [Question 65.1](#) (hset:ns:q:ideals); [Question 23.10](#) is [Question 10.9](#) and Part VIII’s [Question 65.6](#) (hset:ns:q:complexity); [Question 23.11](#) is the remainder of [Question 10.10](#); [Question 23.12](#) is [Question 10.12](#) and Part VIII’s [Question 65.12](#) (hset:ns:q:formalization); [Questions 23.6](#) and [23.7](#) meet Part VIII’s [Questions 65.4](#) and [65.8](#) (hset:ns:q:generators, hset:ns:q:evaluators), which this Part answers in part (the notes after [proposition 19.2](#) and [theorem 19.3](#)).

D Conventions and construction data

Notation	Meaning and location
A	An external set of atoms with $ A \geq \kappa$; it is not an internal set of $M_\kappa(A)$.
$\ker(x)$	Atoms occurring in $\text{TC}(\{x\})$, including x itself when x is an atom.
$M_\kappa(A)$	All objects over A whose atom kernels have cardinality below κ . Pure ranks are unrestricted.
j_f	The recursively lifted atom injection f , fixing every pure object.
B_t	All atoms in all components of type t , not a single component.
D_κ	Union of the entire type blocks that are internal atom sets. It is always an external set but need not be internal.
$\tau(G)$	Number of conjugacy classes of all subgroups of G .
$\tau_{\text{fin}}(G)$	Number of conjugacy classes of finite-index subgroups of G .
$E(g, x)$	One binary evaluator for the whole group action, totalized by the identity outside the pure set G .
$\mathcal{K}(\text{dcl}(\bar{a}))$	Scheme notation for kernels of uniquely definable objects, as explained before theorem 17.1 ; not an added object-language truth predicate.

D.1 An explicit singular-cutoff witness

At $\kappa = \aleph_\omega$, choose disjoint atom blocks B_n of size $\aleph_n$. Partition B_n into cycles of length $n + 2$. Let p rotate each cycle and fix any unused atoms. The single formula

$$“y \text{ is the set of all atoms of exact } p\text{-period } n + 2”$$

has the unique internal value B_n at each $n \in \omega$. The statement of exact period uses finite path graphs and the requirement that no shorter positive iterate returns to the atom. Its range cannot be an internal set because its kernel would contain $\bigcup_n B_n$, of size $\aleph_\omega$. Adding finitely many identity names gives commuting counterexamples of every finite positive arity.

D.2 An explicit regular-cutoff witness

At $\kappa = \aleph_1$, choose distinct staircase codes S_α for $\alpha < \omega_1$ and use countably many copies of each U_{S_α} . Such a family of codes exists by Choice and $\aleph_1 \leq \mathfrak{c}$. Each entire type block is countable. Their union has size $\aleph_1$.

The pure sequence $\langle S_\alpha : \alpha < \omega_1 \rangle$ is an internal parameter, even though it has uncountable size. Its type-block selector is functional on the internal pure domain ω_1 and has no internal range. Each unary coordinate map has just the ray type and preserves Replacement. This explicitly illustrates why large pure input sets cannot be ignored in a model that restricts only atom kernels.

D.3 The finite-cutoff uniformity witness

Take $G = \bigoplus_\omega C_2$ and nonzero characters $\chi_{\{n\}}$ for $n < \omega$. Give each one exactly one two-atom orbit. Separate unary names preserve Replacement, since any finite set of them acts through a finite subgroup. With uniform evaluation, the formula comparing an atom's stabilizer with $\ker \chi_{\{n\}}$ defines the complete two-atom block. Collecting those blocks would require infinitely many atoms, which is forbidden at $\kappa = \omega$. The evaluator's restriction to any internal set still uses only finitely many atoms, so it remains amenable.

E Source and verification record

The following record makes the research boundary reproducible without relying on an unpinned repository branch.

- The target question and prior profile theorem were read in the full source of *Naming Elementary Embeddings* at commit a9ab9a69872c2fb7464097af32fd7b76c75cb030, file SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings/article.tex. Its Git blob is 5426fc8da5f470efc448c3dff041751d0d7dc4a9. The question is Question 10.4, and the profile theorem is Theorem 5.1.
- The group-action connection to Glazer was checked against the full arXiv HTML of Glazer–Yao, version 1, 25 February 2026, especially its kernel-model discussion. LinkedIn identifies the researcher but is not used as the mathematical authority for the results.
- Smith's published 1995 PDF was read directly for the existence of an infinite finitely generated group with three conjugacy classes of subgroups. The construction itself is not reproduced. The original Ivanov–Ol'shanskii chapter is included bibliographically; its publisher metadata was checked, but its paywalled proof was not used.
- The main mathematical arguments received an independent review of the group types, mixed normal forms, strict staircase rigidity, the finite-cutoff definable-kernel argument, and the uniform-action construction. This is an additional review step within preparation of the manuscript, not external peer review.
- The exact finite checks are recorded in `verification_output.txt`. The source and scripts are supplied so that arithmetic and figure generation can be reproduced. The semantic and infinite-cardinal conclusions remain mathematical proofs in the manuscript.

The literature checks support a direct research intersection and a concrete answer to the repository question. They do not establish that every proposed refinement is absent from all unpublished notes or the wider literature. Any submission for publication should retain the prior-work distinctions in [section 12](#) and undergo an additional specialist review.

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